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Reasoning on Structured Data

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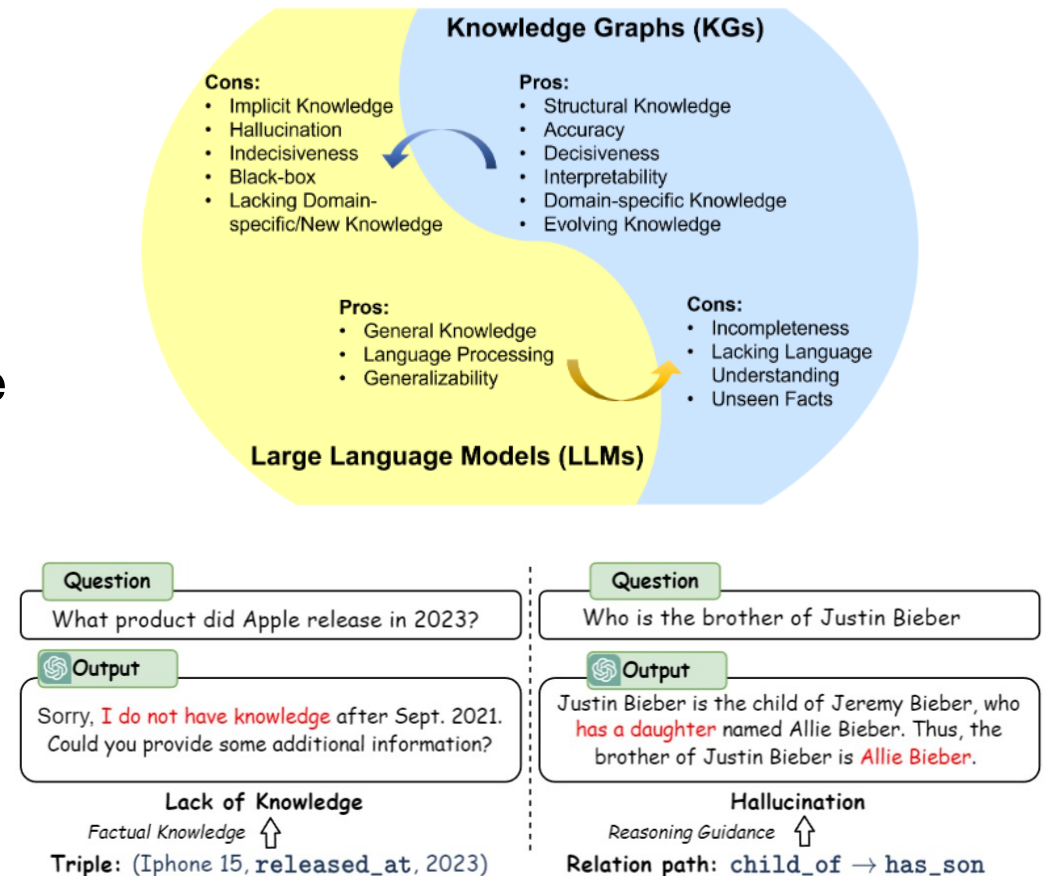
Contents

LLM Reasoning on Graph

- LLM Reasoning on Table

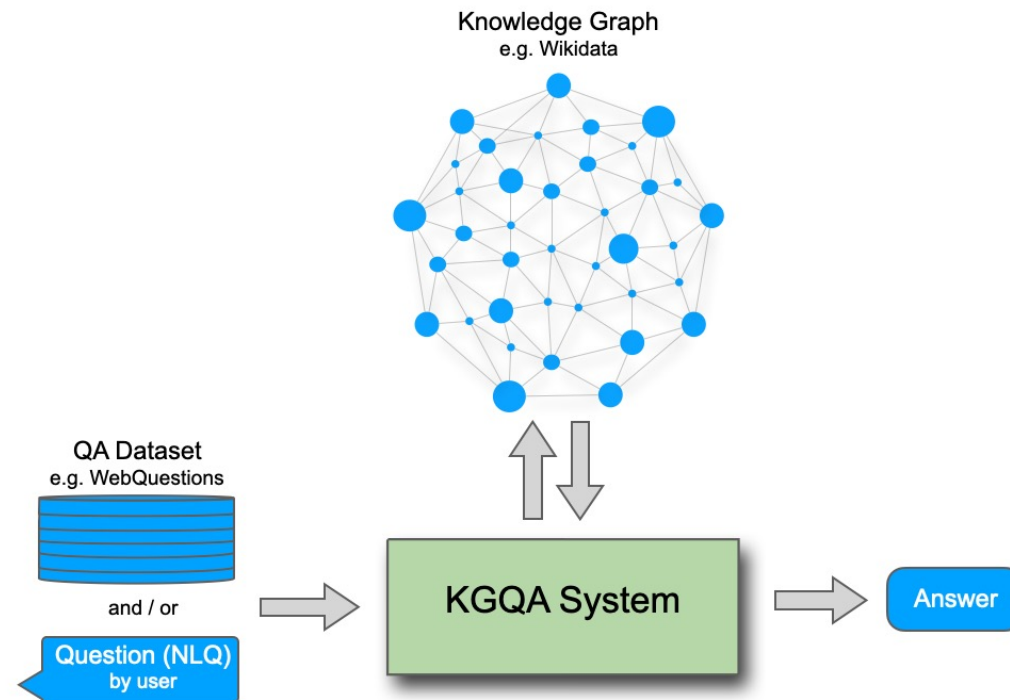
LLMs with Knowledge Graph

- How can KGs assist LLMs?
- LLMs could response with hallucinations
 - Knowledge from KGs mitigate unfaithfulness
 - Even LLMs can provide sources for users
- Internal knowledge in LLMs may be out of date
 - Fine-tuning or unlearning LLMs are laborious
 - Giving the newest information from KGs can solve this
 - Updating KGs are relatively easier



Knowledge Graph Question Answering

- In knowledge graph question answering (KGQA) task, a model should give an answer with a given question where the answer is included in a knowledge graph
 - How effectively can KGs support LLMs?

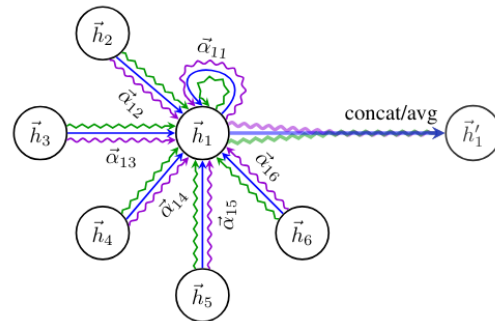


Challenges in KGQA

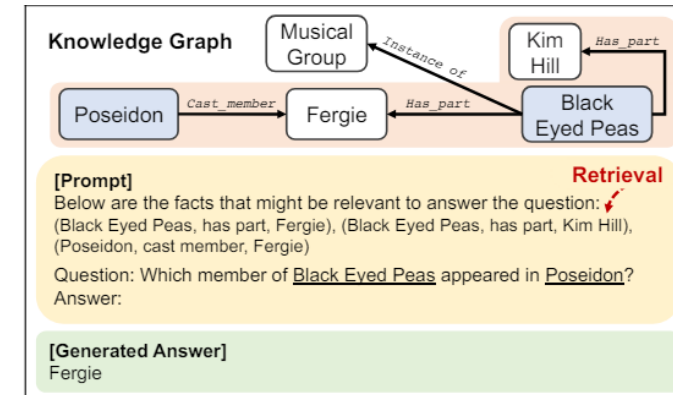
- There exists a gap between text and graph representations
 - Setting additional modules to blend two representations → “**LLMs along with GNNs**”
- LLMs has an ability to reason to some extent but KG itself cannot provide relevant information
 - Exploiting LLMs to search reasoning paths → “**Graph reasoning with LLMs**”
- Above two methods cannot promise the valid reasoning path or induce intensive computations
 - Transforming KG knowledge to make LLMs consume → “**RAG with KG**”

Approaches for LLMs with KG

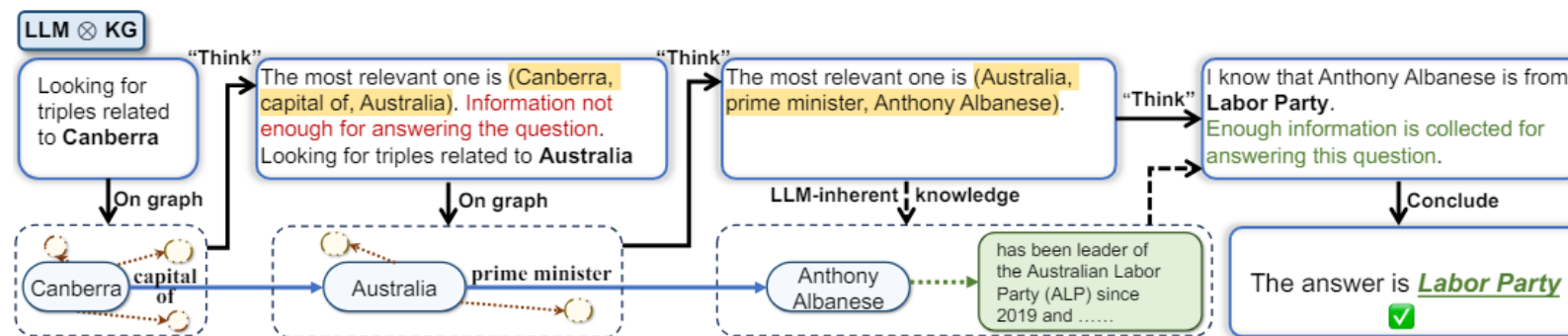
- LLMs along with GNNs
 - Graph embeddings are aligned with text embeddings



- RAG with KG



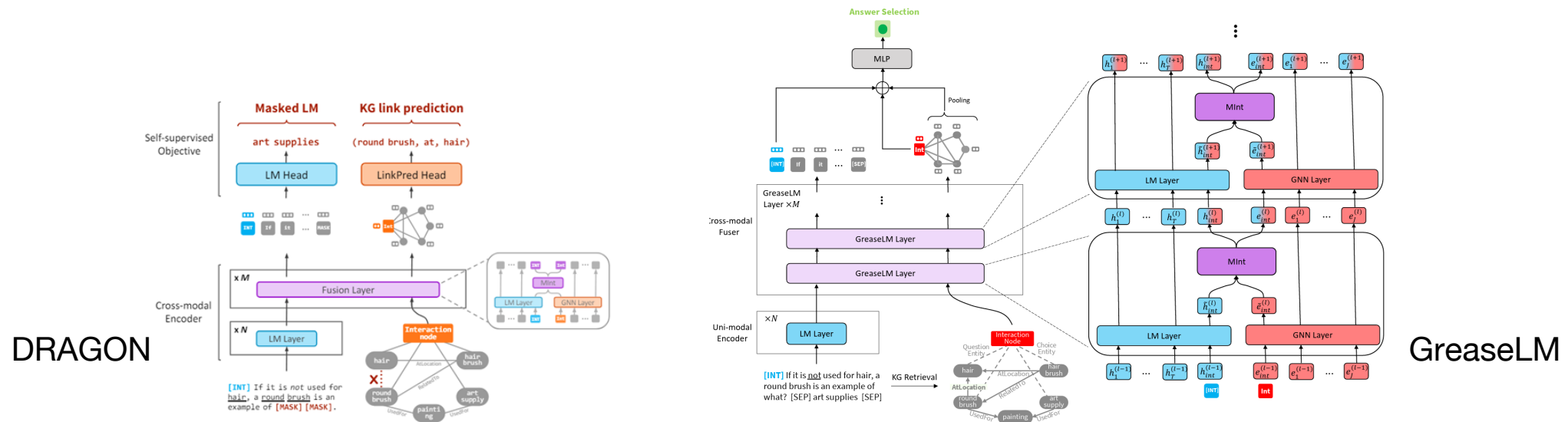
- Graph reasoning with LLMs



- Veličković et al., Graph Attention Networks, ICLR 2018
- Sun et al., Think-on-Graph: Deep and Responsible Reasoning of Large Language Model on Knowledge Graph, ICLR 2024
- Baek et al., Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering, NLRSE 2023

LLMs along with GNNs

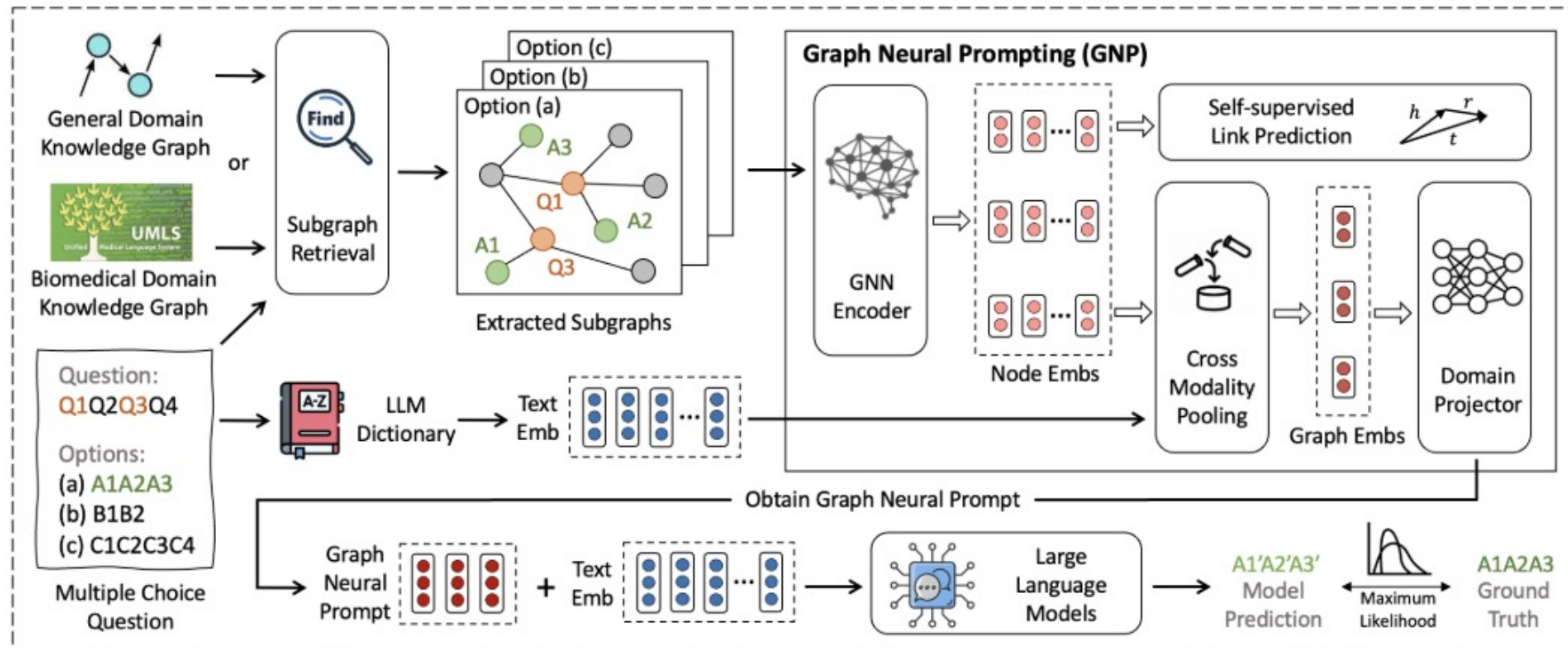
- Recent studies optimize additional modules to blend graph and text representations
 - DRAGON jointly optimizes a language-knowledge model in a self-supervised manner
 - GreaseLM proposes modality interaction (MInt) unit to mix graph and text representations
 - GNP utilizes GNN then concatenate learnable projected graph tokens to text embeddings



- 1) Yasunaga et al., Deep Bidirectional Language-Knowledge Graph Pretraining, NeurIPS 2022
- 2) Zhang et al., GreaseLM: Graph REASONing Enhanced Language Models for Question Answering, ICLR 2022
- 3) Tian et al., Graph Neural Prompting with Large Language Models, AAAI 2024

LLMs along with GNNs: GNP

- Graph Neural Prompting (GNP) utilize GNNs to obtain graph embeddings
- Then, it concatenates learnable projected graph tokens to text embeddings



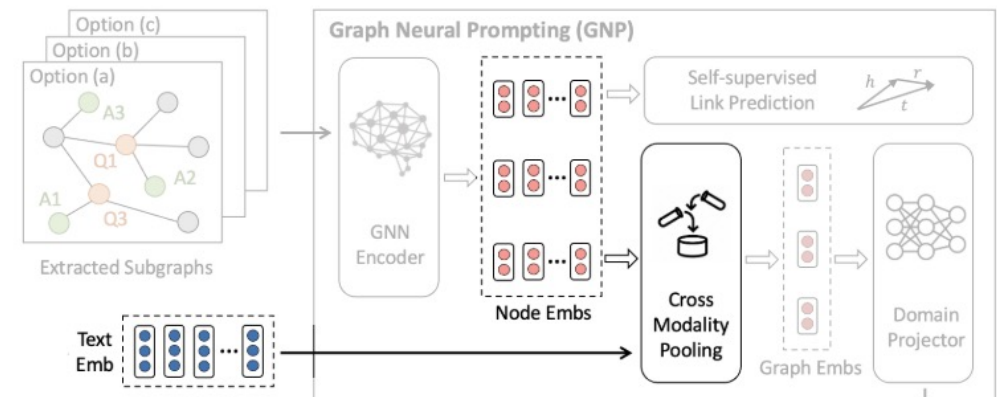
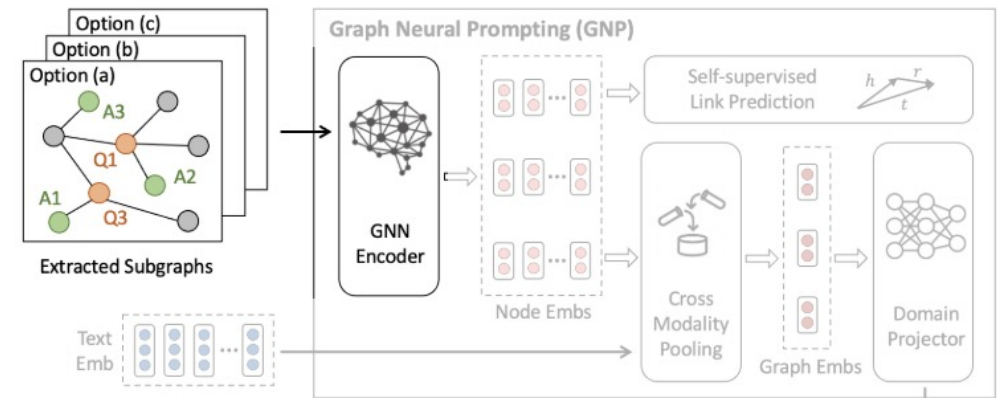
LLMs along with GNNs: GNP

- **GNN Encoder**

- Not all entities in subgraph are relevant to the question
 - Utilize pre-trained GNN (Graph Attention Network) to excavate more relevant knowledge
 - Node embeddings from subgraph $\mathcal{G}'$

- **Cross Modality Pooling**

- Embeddings need to be consolidated to graph-level representation
 - Node: Self-Attention
 - Text: Feed-Forward Network
 - Cross-attention for nodes and texts, then pool



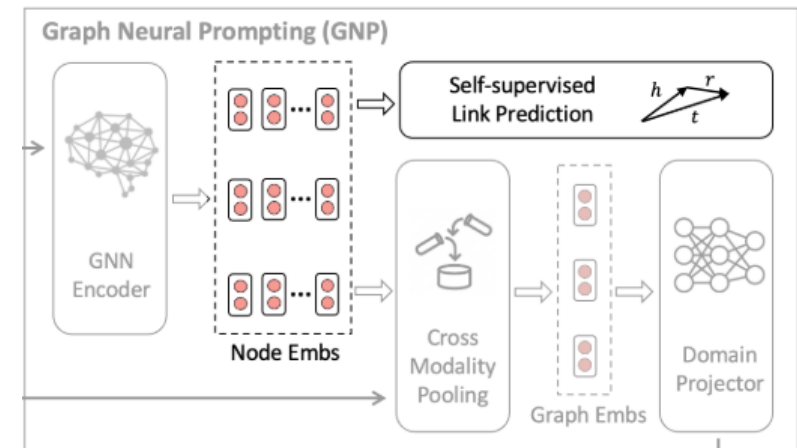
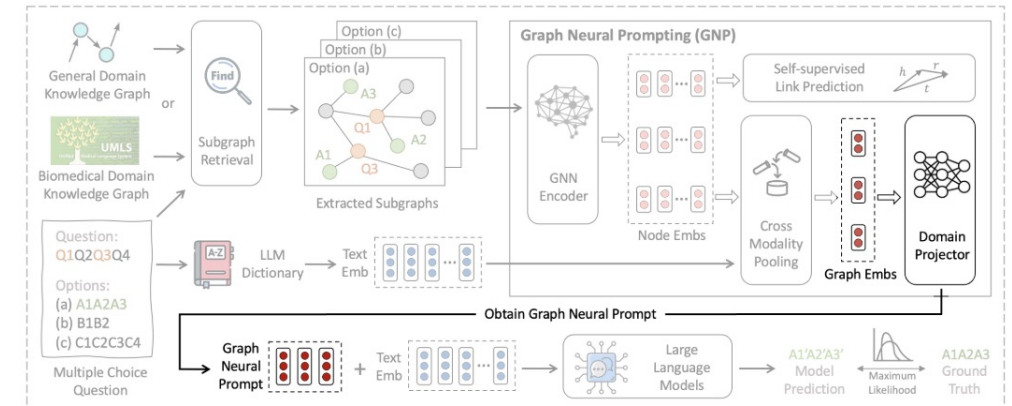
LLMs along with GNNs: GNP

- **Domain Projector**

- It maps graph-level embeddings to LLM tokens
- A simple feed-forward network is trained to bridge disparities between the graph and text

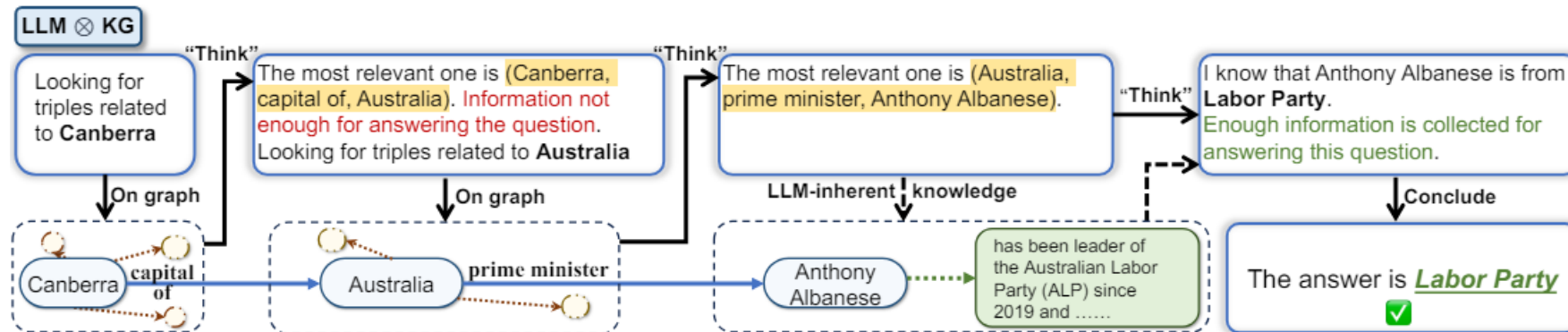
- **Self-supervised Link Prediction**

- Graph augmentation is applied to enhance model comprehension of relationships
 - Edges are masked out
 - Node embeddings and relations are mapped to graph embedding (DisMult)
 - The model should predict the masked edges with trilinear dot product scores of triples



Graph Reasoning with LLMs

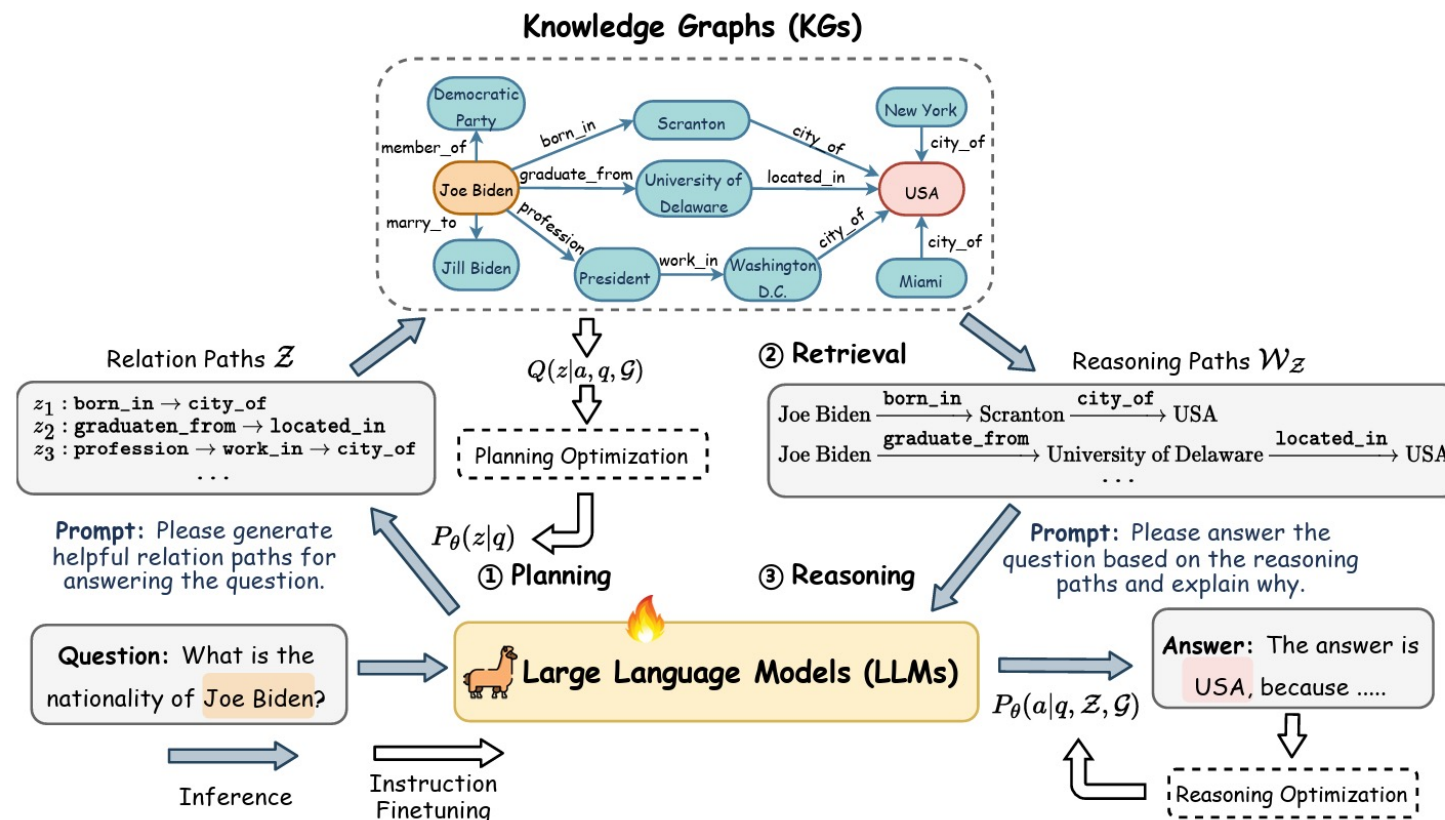
- LLMs can reason on graphs sequentially with heuristics
 - StructGPT queries a KG for each hop in recursive manner
 - ToG queries a KG in CoT manner
 - Talk Like A Graph encodes graph-structured data as text for LLMs
 - RoG finetunes LLMs to find useful reasoning paths



- 1) Jiang et al., StructGPT: A General Framework for Large Language Model to Reason over Structured Data, EMNLP 2023
- 2) Sun et al., Think-on-Graph: Deep and Responsible Reasoning of Large Language Model on Knowledge Graph, ICLR 2024
- 3) Fatemi et al., Talk like a Graph: Encoding Graphs for Large Language Models, ICLR 2024
- 4) Luo et al., Reasoning on Graphs: Faithful and Interpretable Large Language Model Reasoning, ICLR 2024

Graph Reasoning with LLMs: RoG

- Reasoning on Graph (RoG) proposes a “Planning-Retrieval-Reasoning” framework



Graph Reasoning with LLMs: RoG

- An LLM is optimized to predict the best answer (a) with a given question (q) and KG ($\mathcal{G}$)
 - This objective can be bounded by ELBO

$$\boxed{\log P(a|q, \mathcal{G})} \geq \mathbb{E}_{z \sim Q(z)} [\log \boxed{P_{\theta}(a|q, z, \mathcal{G})}] - D_{\text{KL}}(Q(z) \parallel \boxed{P_{\theta}(z|q)})$$

Retrieval-Reasoning **Planning**

- Maximizing the expectation of the log probability of predicting the answer along with relation path (z)
- Minimizing the KL-divergence of between the posterior and the prior of faithful relation paths

Graph Reasoning with LLMs: RoG

- “**Planning**”-Retrieval-Reasoning framework

- RoG distills the knowledge from KGs into LLMs to **generate faithful relation paths** (z) as plans

- Example of relation path (z)

- Question: Who is the child of Alice?

$$z = \text{marry_to} \rightarrow \text{father_of}$$
$$w_z = \text{Alice} \xrightarrow{\text{marry_to}} \text{Bob} \xrightarrow{\text{father_of}} \text{Charlie}$$

- Posterior can be approximated as below

- $Q(z) \simeq Q(z|a, q, \mathcal{G}) = \begin{cases} \frac{1}{|\mathcal{Z}|}, \exists w_z(e_q, e_a) \in \mathcal{G}, \\ 0, \text{else}, \end{cases}$

- Only path that connects the question entity (e_q) and the answer entity (e_a)
 - Assume that all valid relation paths ($\mathcal{Z}$) are uniformly distributed

- The planning loss can be derived as below

$$\mathcal{L}_{\text{plan}} = D_{\text{KL}}(Q(z) \| P_{\theta}(z|q)) = D_{\text{KL}}(Q(z|a, q, \mathcal{G}) \| P_{\theta}(z|q)) \simeq -\frac{1}{|\mathcal{Z}^*|} \sum_{z \in \mathcal{Z}^*} \log P_{\theta}(z|q),$$

Graph Reasoning with LLMs: RoG

- Planning-“**Retrieval-Reasoning**” framework
 - RoG enable LLMs to **conduct reasoning** based on **the retrieved reasoning paths**
 - The expectation is approximated with K sampled plans $\mathcal{Z}_K^*$

$$\mathcal{L}_{\text{reason}} = \mathbb{E}_{z \sim Q(z|a, q, \mathcal{G})} [\log P_{\theta}(a|q, z, \mathcal{G})] = \sum_{z \in \mathcal{Z}_K^*} \log P_{\theta}(a|q, z, \mathcal{G}) = \log P_{\theta}(a|q, \mathcal{Z}_K^*, \mathcal{G})$$

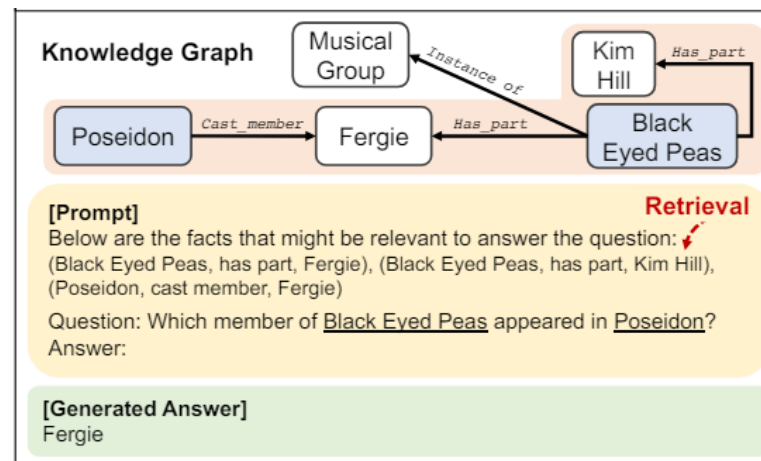
- Retrieval process
 - Given a question (q) and relation path (z), do BFS on KG ($\mathcal{G}$) to retrieve reasoning path $\mathcal{W}_z$

$$\mathcal{W}_z = \{w_z(e_q, e_*) | w_z(e_q, e_*) = e_q \xrightarrow{r_1} e_1 \xrightarrow{r_2} \dots \xrightarrow{r_l} e_{a*}, w_z(e_q, e_*) \in \mathcal{G}\}$$

- Reasoning process
 - Instruction fine-tune with a template

RAG with KG

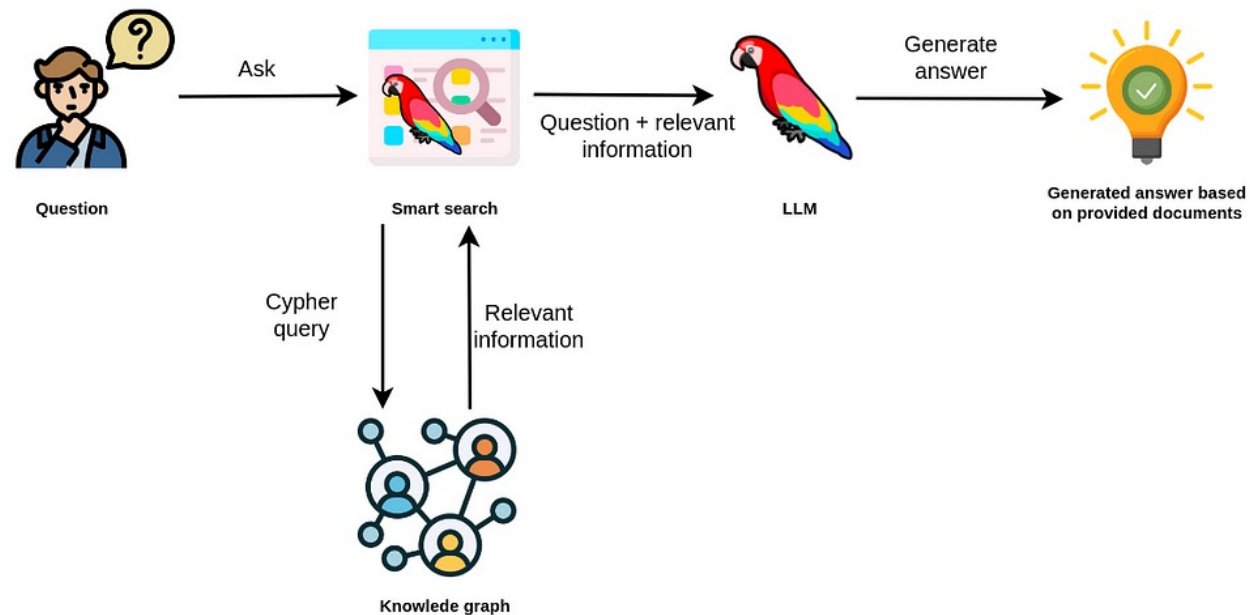
- KG information is retrieved and verbalized into texts for LLMs
 - KAPING linearizes triples into tuple-form
 - Retrieve-Rewrite-Answer verbalizes each path into free-form text
 - EFSum summarizes triples with preference alignment



- 1) Baek et al., Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering, NLRSE 2023
- 2) Wu et al., Retrieve-Rewrite-Answer: A KG-to-Text Enhanced LLMs Framework for Knowledge Graph Question Answering, IJCKG 2023
- 3) Ko et al., Evidence-Focused Fact Summarization for Knowledge-Augmented Zero-Shot Question Answering, EMNLP 2024

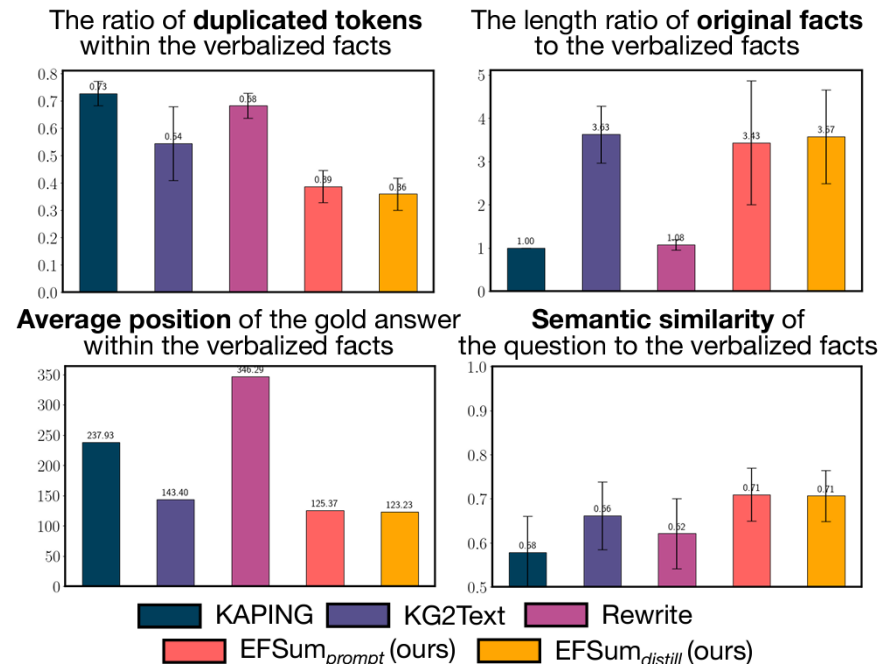
RAG with KG

- Retrieval-Augmented Generation (RAG) with KG
 - Fact Retrieval: Return the knowledge (subgraph, triples...) from a KG with a given question
 - Fact Verbalization: Align text-form question with retrieved knowledge
 - Fact Injection: Predict the answer given the augmented prompt



RAG with KG: EFSum

- Existing fact verbalization approaches suffer from
 - Low density of evidence:** Duplicated tokens ($\uparrow$), Compression ratio ($\downarrow$)
 - Low clarity of evidence:** Answer position ($\uparrow$), Question-knowledge similarity ($\downarrow$)



Question: Where was george washington carver from?
Answer: Diamond

KAPING: (George Washington Carver, place of birth, Diamond),(George Washington Carver, given name, George),(George Washington Carver, occupation, biologist),(George Washington Carver, interested in, botany),(George Washington Carver, occupation, university teacher),(George Washington Carver, field of work, pedagogy),(George Washington Carver, relative, Moses Carver),(George Washington Carver, place of birth, United States of America),(George Washington Carver, residence, Tuskegee),(George Washington Carver, occupation, inventor)

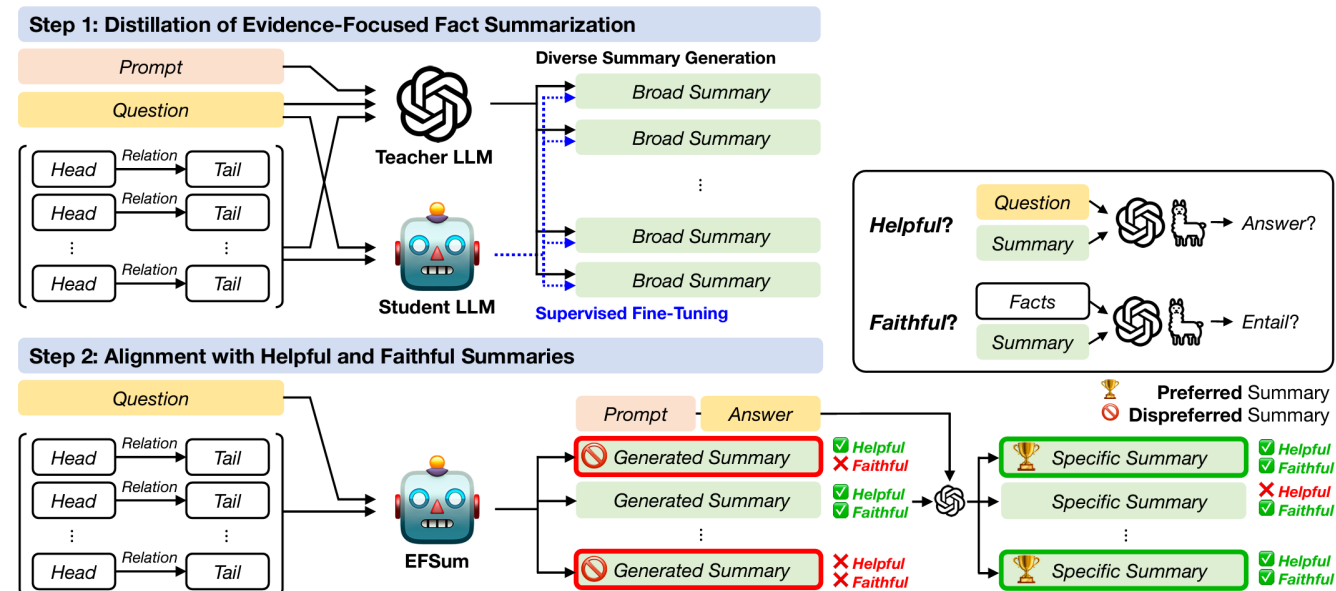
Rewrite: George Washington Carver is associated with the Commons category "George Washington Carver". George Washington Carver was a botanist. George Washington Carver was an inventor. George Washington Carver was a university teacher. George Washington Carver was a mycologist. George Washington Carver was a chemist. George Washington Carver was a biologist. George Washington Carver was a citizen of the United States of America. George Washington Carver died on January 5, 1943. George Washington Carver is a human.

EFSum_{distill}: George Washington Carver, the renowned biologist, **was born in Diamond**, United States of America. He took great interest in botany and became a university teacher, focusing on pedagogy. Additionally, Carver contributed significantly as an inventor. His relative, Moses Carver, also had a close association with him. Carver later resided in Tuskegee.

High density
High clarity

RAG with KG: EFSum

- EFSum presents a framework for evidence-focused fact summarization
- The distilled EFSum model adopts preference alignment for summary generation
 - **Helpfulness:** Can LLMs correctly answer the question based on the summary?
 - **Faithfulness:** Is the summary factually consistent with the provided set of facts?

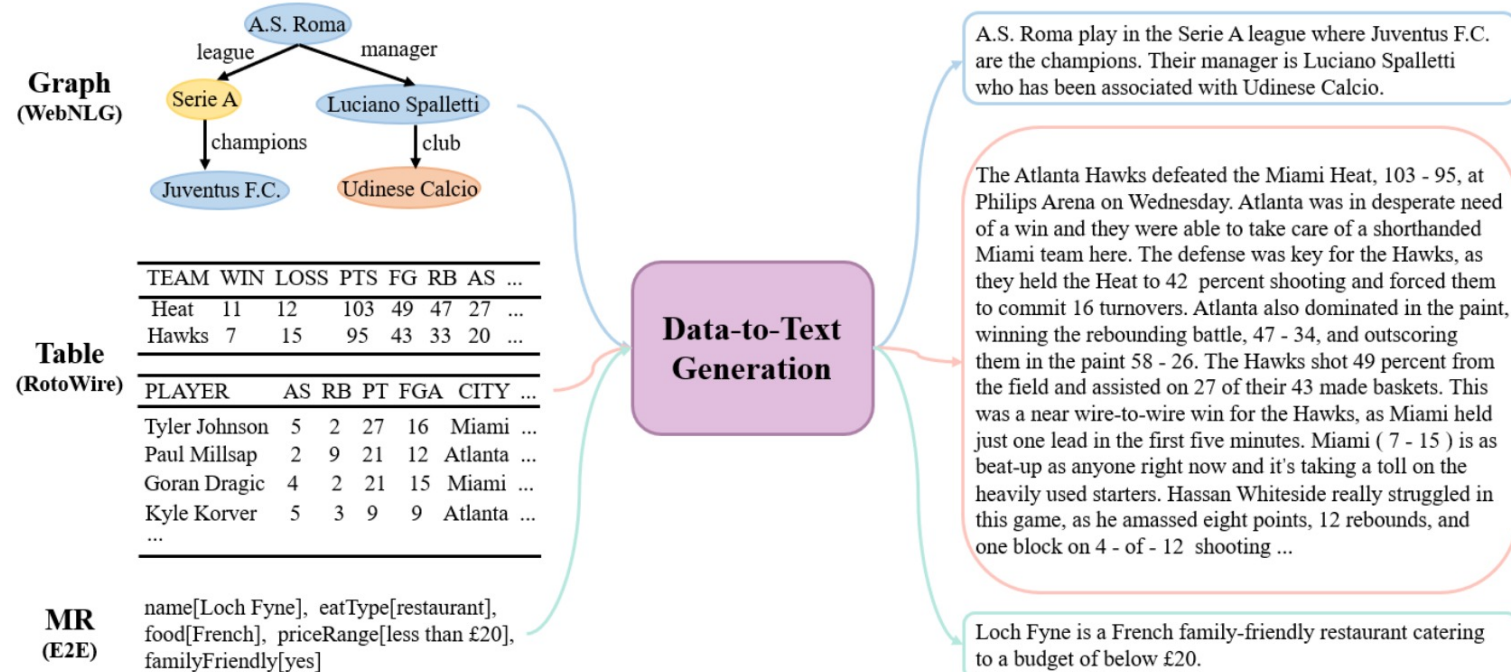


Contents

- LLM Reasoning on Graph
- ✓ **LLM Reasoning on Table**

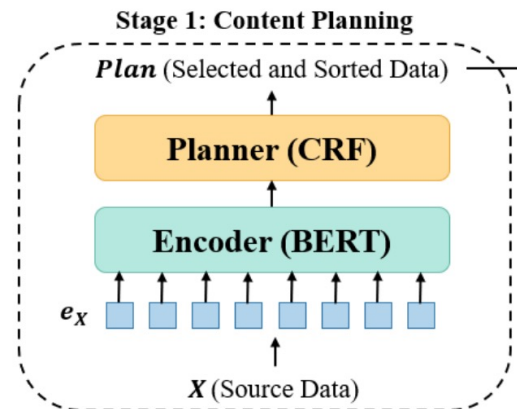
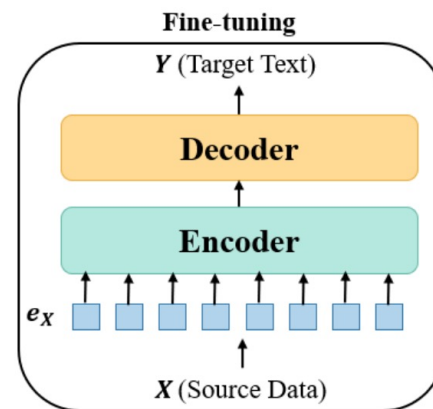
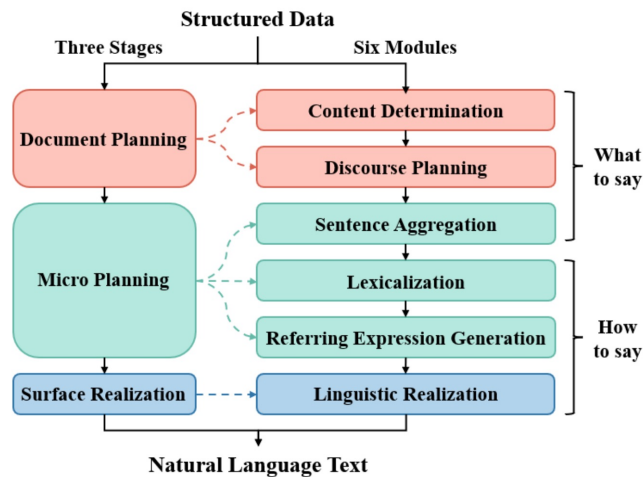
Data-to-Text Generation

- Data-to-Text (D2T) generation refers to the task of **generating natural language from structured data** such as graphs, tables and meaning representations
 - Natural language that accurately and fluently describes structured data assists people in understanding data efficiently and easily

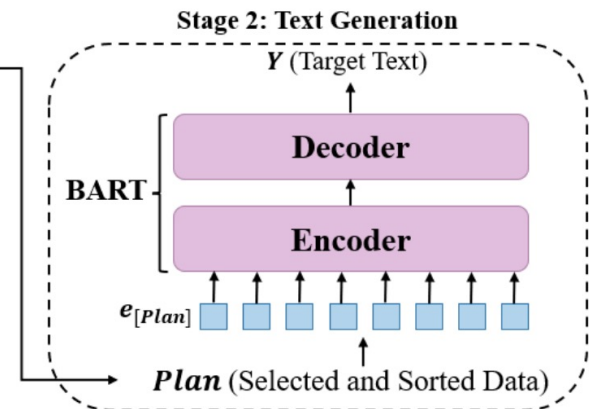


Data-to-Text Generation

- D2T aims to generate the textual output that can accurately and fluently describe the non-linguistic structured data
 - **What to Say?** which part of the given data is selected for generation
 - **How to Say?** accurately / fluently / diversely/ .. etc
- Traditional vs. Neural Approach



What to say?



How to say?

Table Representation Learning

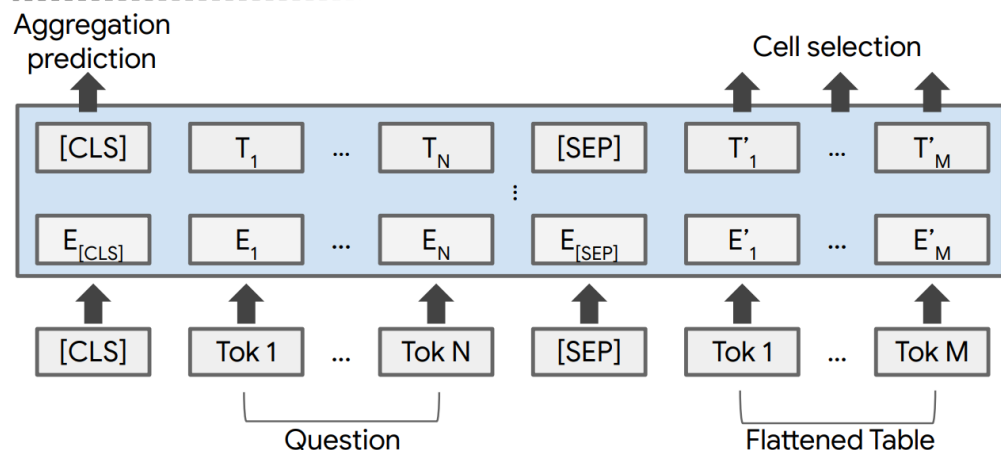
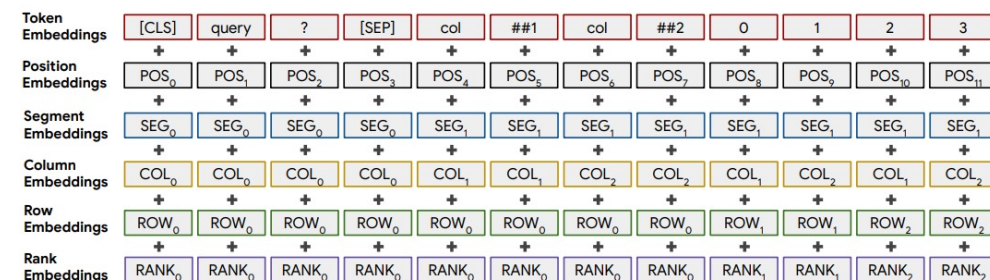
• TAPAS

op	$P_a(op)$	compute(op, P_s, T)
NONE	0	-
COUNT	0.1	.9 + .9 + .2 = 2
SUM	0.8	.9 × 37 + .9 × 31 + .2 × 15 = 64.2
AVG	0.1	64.2 ÷ 2 = 32.1

$$s_{pred} = .1 \times 2 + .8 \times 64.2 + .1 \times 32.1 = 54.8$$

Rank	...	Days	P_s
1	...	37	0.9
2	...	31	0.9
3	...	17	0
4	...	15	0.2
...	...	...	0

Table	
col1	col2
0	1
2	3



- **Table-aware embeddings** – Every token = WordPiece + Pos + Column ID + Row ID (+ Rank for numbers) so the model “knows” same-row vs. same-column relations.
- **Input step** – Flatten the table row-wise, prepend the question, add those embeddings, cap at 512 tokens.
- **Pre-train** – Standard MLM/NSP **plus** two table tasks: (i) predict answer-cells, (ii) predict required aggregation (NONE / SUM / COUNT / AVG).
- **Fine-tune & Infer** – Tag answer cells and choose an aggregator; at test time return the tagged cell texts or apply the chosen SUM/COUNT/AVG.

Table Representation Learning

• MATE

Row and Column related Attention Masking

Query: Which college got the most players?

Pick #	Player	College
27	Tom	Wilfrid Laurier
28	Mark	York
29	Pedro	California
30	Rob	York
...	...	...

Row Head

Column Head

- in order to create better cell representation, attending to same column, row are important!

$$A_k^i = \text{AttentionPattern}$$

$$Q = \{[cls], QueryText, [sep]\}$$

$$A_k^i = \begin{cases} \{1, \dots, n\} & k \in Q, \text{ else} \\ Q \cup \{j : r_j = r_k\} & 1 \leq i \leq h_r \\ Q \cup \{j : c_j = c_k\} & \text{otherwise} \end{cases}$$

Query: which college got the most players

Player	College
Tom	Wilfrid
Mark	York

Query: which college got the most players

Player	College
Tom	Wilfrid
Mark	York

row heads Attention Mask

	Query	Player	College	Tom	Wilfrid	Mark	York
Query							
Player				..	..	..	..
College				..	..	..	..
Tom		..	..			..	..
Wilfrid		..	..			..	..
Mark		..	..	..	..		
York		..	..	..	..		

column heads Attention Mask

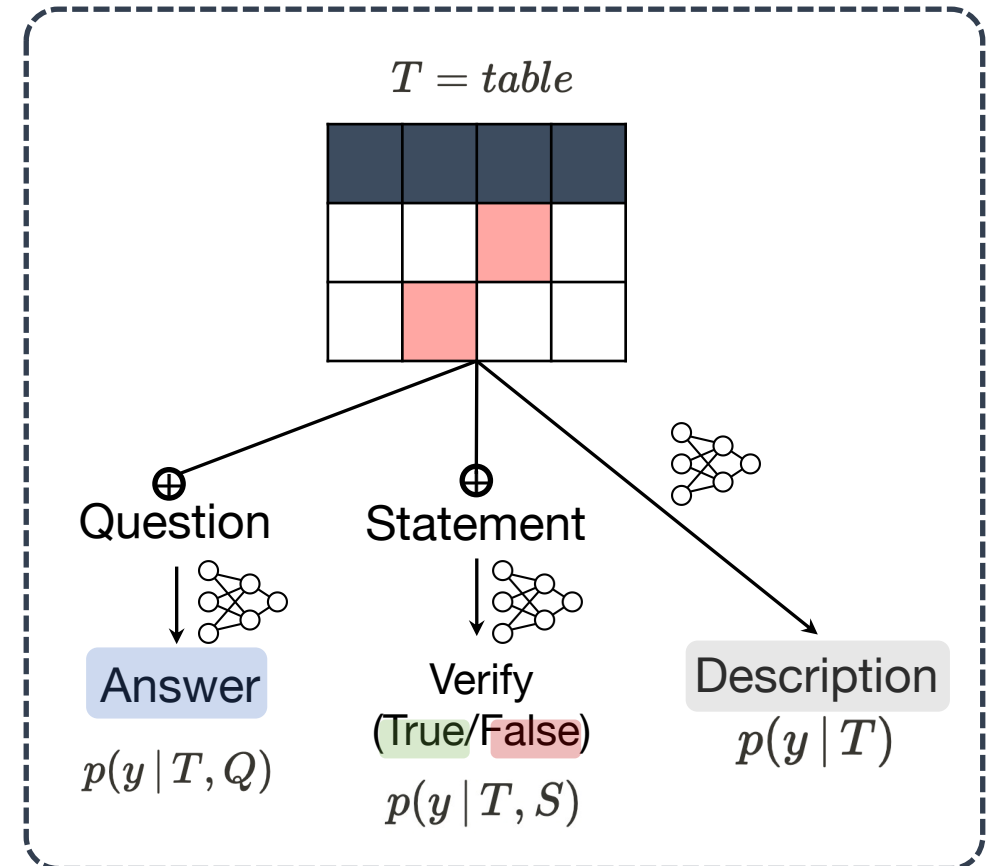
	Query	Player	College	Tom	Wilfrid	Mark	York
Query							
Player			..		..		..
College		..		..		..	
Tom			..		..		..
Wilfrid		..	..	..		..	
Mark			..		..		..
York		..		..		..	

$$\text{AttentionHead}_k^i(\mathbf{X}) = \mathbf{W}_V^i \mathbf{X}_{A_k^i} \text{softmax} | (\mathbf{W}_K^i \mathbf{X}_{A_k^i})^T \mathbf{W}_Q^i \mathbf{X}_k$$

[CLS] [which] [college] [got] ... [?] [SEP] [Pick] [Play] [##er] [College] [27] [Tom] [Wilfrid] [28] [Mark] [York] [29] [Pedro] [California]..

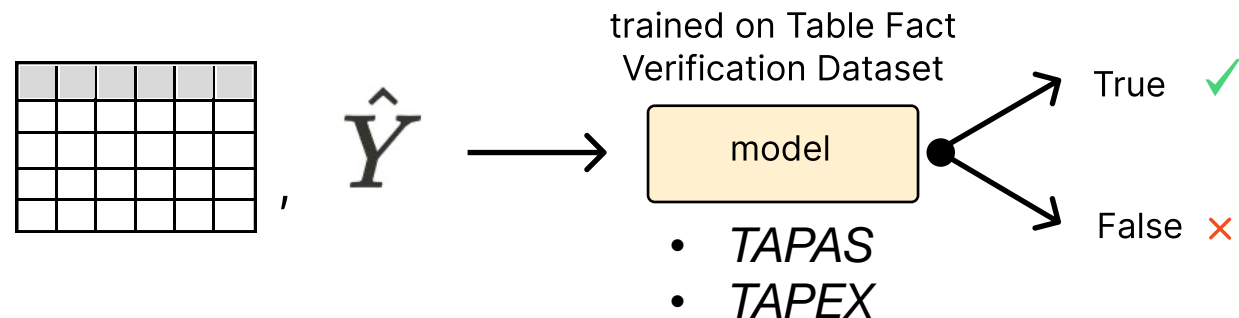
Table Reasoning

- Downstream tasks (Reasoning over tables)
 - Table Fact Verification
 - Classifying whether statement is fact or not (entailed or refuted) *table-to-text generation*
 - Table Question Answering
 - Table + Question $\rightarrow$ Answer (short form answer | long-form answer)
 - Table Description generation
 - Generating meaningful descriptions from table
- Input Serialization (Linearization)
 - col : Season | Team | Goals | ... row 1: 2001 | A.Utd | 2 | ...



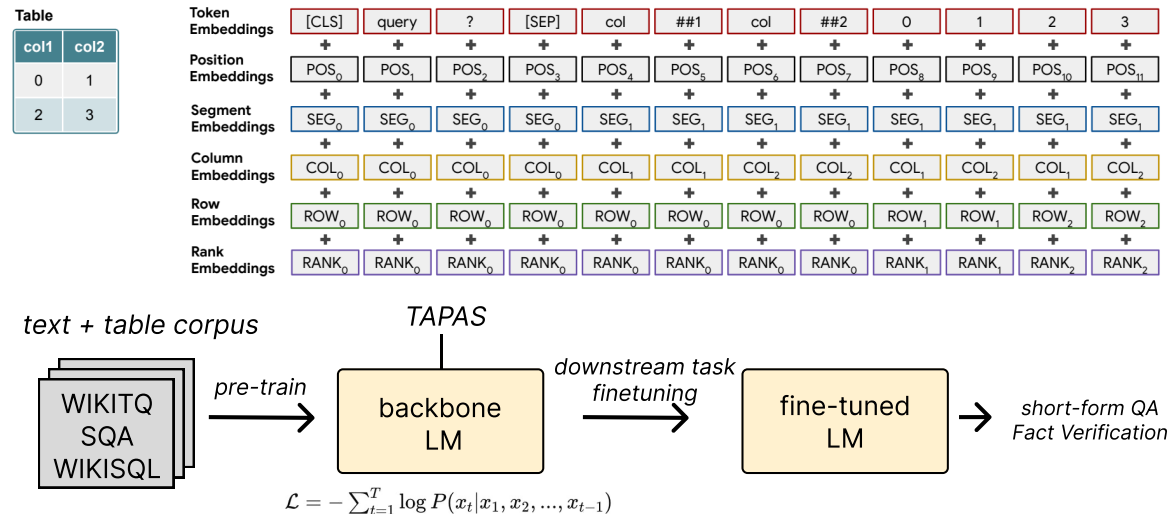
Evaluation on Table Reasoning

- For Fact Verification & short-form QA task
 - Acc, F1
- For long-form QA & description Generation
 - **Surface-level:** BLEU, ROUGE, METEOR, BERTScore, ...
 - Comparing generated text with the target text by computing lexical overlap or semantic similarity
 - **Faithfulness-level:** TAPAS-Acc, TAPEX-Acc
 - Use trained model to evaluate the factual correctness of the generated text



Faithfulness Evaluation: TAPAS

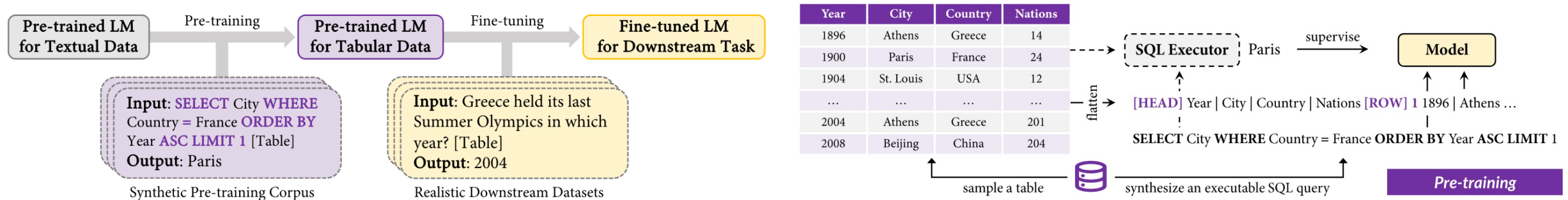
- TAPAS is the **pre-trained foundation model** in table-based tasks
 - It pretrains an LM with massive table-text pairs collected from diverse datasets
 - Generic language modeling objective is used with additional table-aware position embedding
- Downstream task: short-form QA, fact verification
 - TAPAS-Acc use the model trained on TabFACT dataset



1) Herzig et al., TAPAS: Weakly Supervised Table Parsing via Pre-training, ACL 2020

Faithfulness Evaluation: TAPEX

- TAPEX is the **pre-trained foundation model** in table-based tasks
 - It pretrains an LM as a neural SQL executor with synthetic SQL corpus to enhance the table-based task understanding
- Downstream task: short-form QA, fact verification



- TAPEX-Acc use trained model on TabFACT dataset

Diversity of Table-to-Text Generation

- Can we provide **various perspectives (insights)** on the data?
- Can we offer **users more choices**?
 - Without considering diversity, only information from the same/similar table region or logical operation can be covered => can't fully utilize the whole data

2008 Champions Tour

Rank	Player	Country	Earnings	Wins
1	Hale Irwin	United States	24,920,665	45
2	Gil Morgan	United States	18,964,040	25
3	Dana Quigley	United States	14,406,269	11
4	Bruce Fleisher	United States	13,990,356	18
5	Larry Nelson	United States	13,262,808	19

w/o diversity enhance

Hale Irwin and Gil Morgan represent the same country
Hale Irwin had **more** wins than Gil Morgan
Hale Irwin had **more** earnings than Gil Morgan
Hale Irwin had the **more** wins than Bruce Fleisher
Bruce Fleisher had the **highest earnings** of any player with 13,990,356

w diversity enhance

The **average** earnings of Hale Irwin and Dana Quigley is 19,663,467
Five of the players are from the same country, United States
Dana Quigley had **less** wins than Larry Nelson
Most of the players had earnings less than 18,964,040
Larry Nelson had the **least** number of earnings

Challenges of Table Reasoning

- **Joint understanding** of text & structured tabular data
 - Pre-training task formulation
 - Automatic construction of high-qual pre-training corpus
 - How to maximize the reasoning ability of LM on table
- Problems occurred from **large and complex tables**
 - How to encode table information concisely and effectively
 - How to decompose long table to sub-small table
 - How to faithfully state the description without distracted to irrelevant information on table

Table-to-Text Generation

- Table-to-text (T2T) generation aims to produce **natural language descriptions** from structured tables
 - Statement generated from tables can be inferred based on different levels of information
 - Range from a value of a specific cell to the result of logical or numerical operations across multiple cells, such as the average value of a column, or a comparison between rows
- How can we **fluently** and **faithfully** generate texts based on table (factual consistency)?
 - Limitation: existing T2T models mainly focus **on surface-level realization**

Medal Table from Tournament

Nation	Gold Medal	Silver Medal	Bronze Medal	Sports
Canada	3	1	2	Ice Hockey
Mexico	2	3	1	Baseball
Colombia	1	3	0	Roller Skating

Surface-level Generation

Sentence: Canada has got 3 gold medals in the tournament.
Sentence: Mexico got 3 silver medals and 1 bronze medal.

Table-to-Text Generation

- it is challenging for the model to capture information **beyond surface-realization** in an end-to-end approach
- Due to its structured nature, the evidence that should be searched from the table to describe it in the target is **not explicitly stated** in the table cells

Title: Kyle's Football Career Statistics

Season	Team	...	Injury	Goals
2003	A.Utd		X	2
2004	A.Utd		O	0
2005	A.Utd		X	1
2006	A.Utd		O	0
2007	B.FC		O	3
2008	B.FC		X	9
2009	B.FC		X	7
2010	B.FC		O	2
2011	C.City		O	0
2012	C.City		X	3

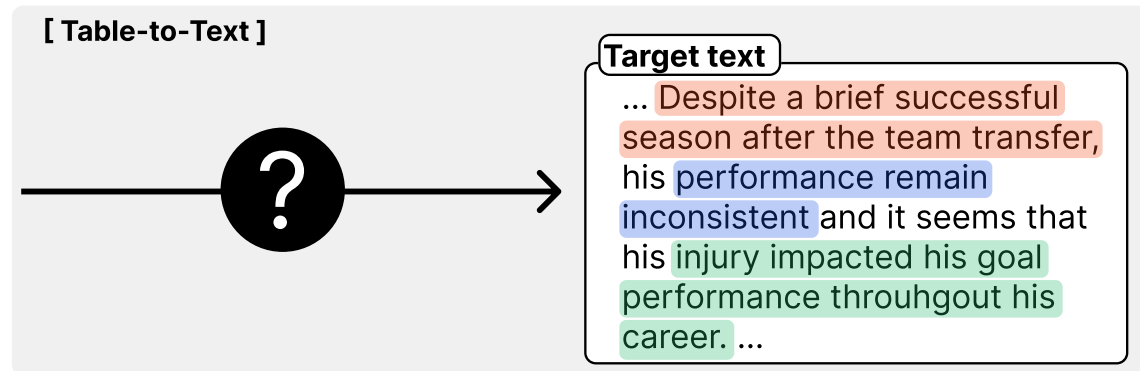
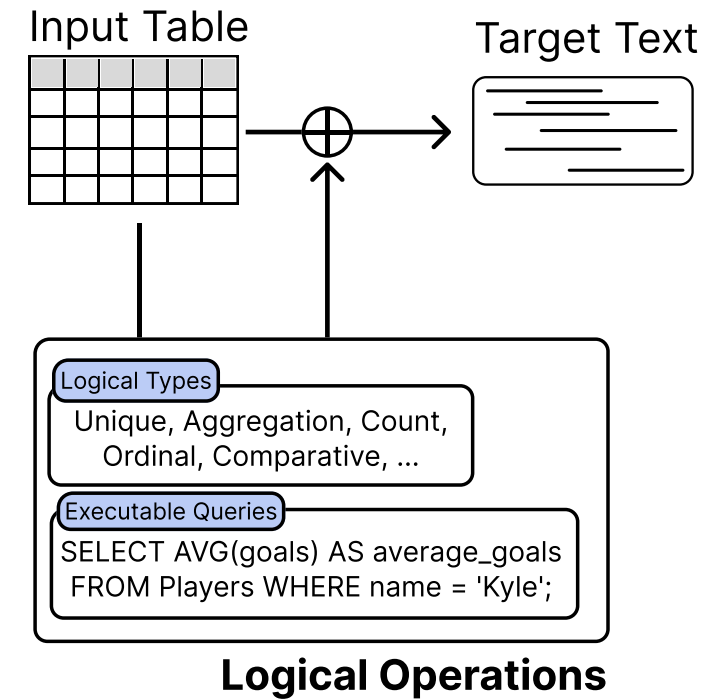


Table-to-Text Generation

- A line of research address the **uncontrollability problem** by **injecting knowledge** as a mediator
 - They deal uncontrollability with knowledge enhancement
- They guide what kind of table contents should be stated in the text description
 - These knowledge include **logical table operations**
- Logical Type
 - Executable languages which mimics SQL



Logical Table-to-Text Generation

- Challenges
 - Factual correctness:** Generated sentence should be faithful to the underlying data
 - Enhance diversity:** Generated sentence should provide diverse aspects of data
 - Deal with trade-off problem:** with only simple decoding techniques, high quality hinders diversity

Medal Table from Tournament				
Nation	Gold Medal	Silver Medal	Bronze Medal	Sports
Canada	3	1	2	Ice Hockey
Mexico	2	3	1	Baseball
Colombia	1	3	0	Roller Skating

Surface-level Generation
Sentence: Canada has got 3 gold medals in the tournament.
Sentence: Mexico got 3 silver medals and 1 bronze medal.

Logical Natural Language Generation
Sentence: Canada obtained 1 more gold medal than Mexico .
Sentence: Canada obtained the most gold medals in the game.

Humans have ability to generalize **beyond superficial facts**, generating statements that can be **logically entailed** by the given data

- logical reasoning = **symbolic operation**
(max, min, sum, more, average..)

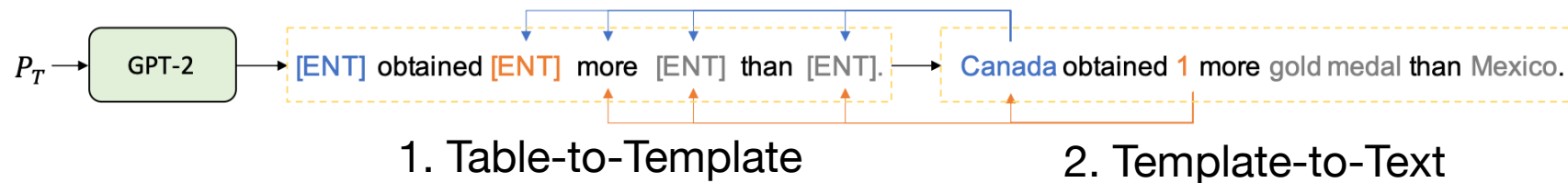
Logical Table-to-Text Generation

- Existing NLG datasets are mainly composed of surface-level description over the given recodes
- LogicNLG involves **rich logical inference**, **open-domain** tables and annotated sentences with minimum domain-knowledge

Medal Table from Tournament				
Nation	Gold Medal	Silver Medal	Bronze Medal	Sports
Canada	3	1	2	Ice Hockey
Mexico	2	3	1	Baseball
Colombia	1	3	0	Roller Skating

Surface-level Generation	
Sentence: Canada has got 3 gold medals in the tournament.	
Sentence: Mexico got 3 silver medals and 1 bronze medal.	

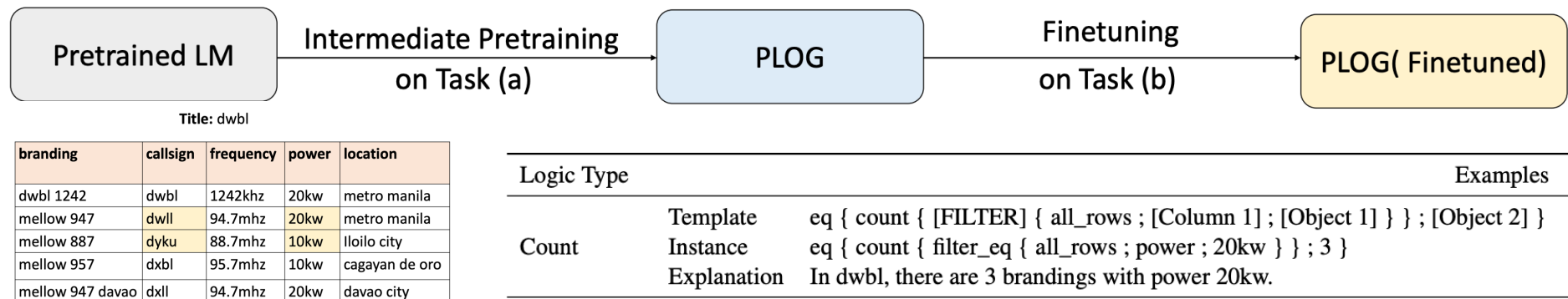
Logical Natural Language Generation	
Sentence: Canada obtained 1 more gold medal than Mexico.	
Sentence: Canada obtained the most gold medals in the game.	



$$p(\tilde{Y}|T), \tilde{Y} = [Y_{template}; [sep]; Y]$$

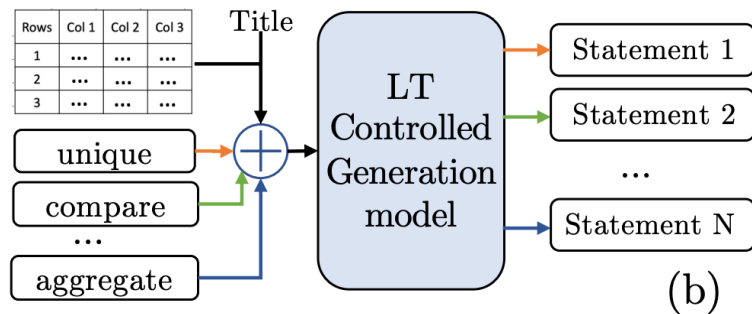
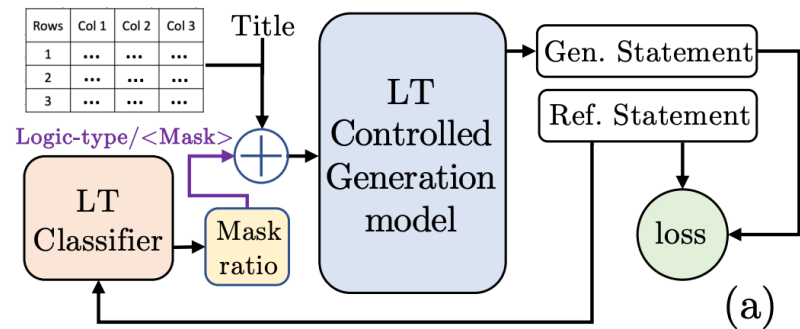
Logical Table-to-Text Generation

- PLOG is first pretrained on a table-to-logical-form generation (table-to-logic) task, then finetuned on downstream table-to-text tasks.
 - **Task (a):** Pretrain Table-to-Logic
 - Expect the model to learn the abstract content planning via unambiguous logical forms
 - {Table} -> {logical_form 1,2,3...}
 - **Task (b):** Fine-tune with numeric pre-computation
 - Append rank of each numerical cells to linearized Table



Logical Table-to-Text Generation

DEVTC



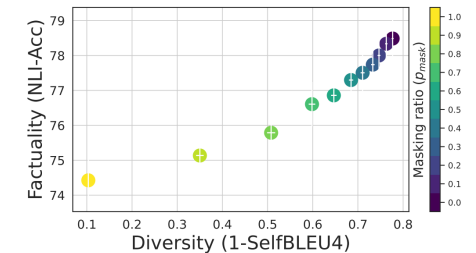
LT Classifier

- Using Logic-2-Text dataset , fine-tune BERT (annotate LogicNLG with this BERT {Table, reference, logical_type})
- Concat logical type with linearized Table

$$\mathbf{y} = \operatorname{argmax} \prod_{i=1}^N P(y_i | y_{<i}, T, lt; \theta) \quad lt = \text{logical type}$$

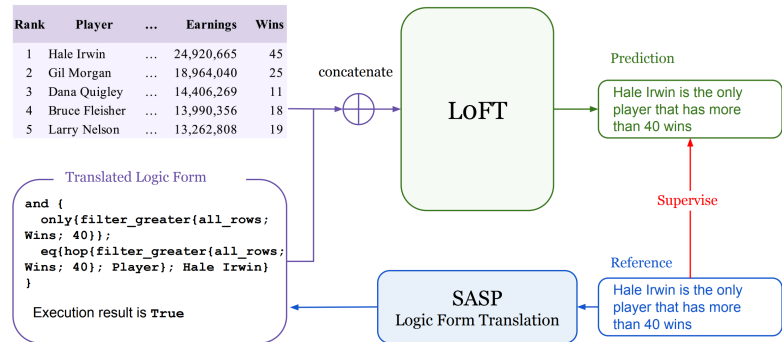
Masking

- model learn how to condition on LT,
- also enable robustness for scenario where LT is unavailable

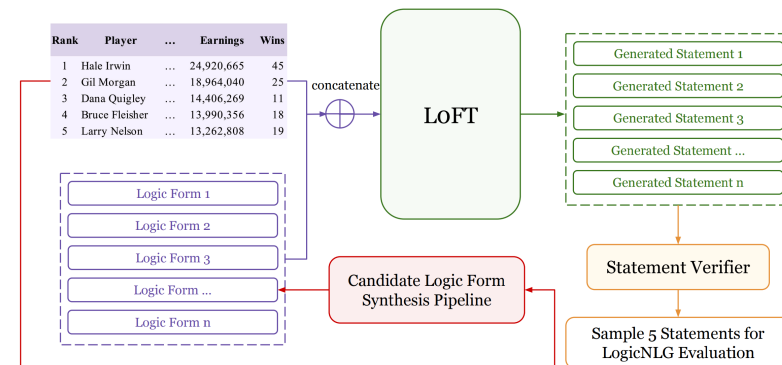


Logical Table-to-Text Generation

LoFT



(a) LoFT training stage.



(b) LoFT inference stage.

Train

- construct {statement, logic form} parallel data for the LogicNLG training set using SASP

$$\mathbf{y} = \operatorname{argmax} \prod_{i=1}^N P(y_i | y_{<i}, T, l; \theta),$$

$T = \text{linearized table}, l = \text{logical form}$

Inference

- Logic Form Synthesis Pipeline (Content Planner)
 - make 45 different templates based on training data (use PLOG, Table-to-LogicForm)
 - categorize by 8 abstract form, execute each on table, collect True
- Statement Verifier
 - NLI-based verifier filter out unfaithful generations (Train RoBERTa with TABFACT)
 - candidate statements -> filter NLI entailment with Table=True

LLM-based Table-to-Text Generation

- More challenging datasets for table-to-text generation **require intensive-reasoning** to generate long-form paragraph from table input

Title: United States House of Representatives Elections, 2012

District	Incumbent	First Elected	Party	Candidates
North Carolina 3	Walter Jones Jr	1994	Republican	Walter Jones Jr (R) 63.2% Erik Anderson (D) 36.8%
North Carolina 4	David Price	1996	Democratic	David Price (D) 74.4% Tim D'Annunzio (R) 25.6%
North Carolina 6	Howard Coble	1984	Republican	Howard Coble (R) 60.9% Tony Foriest (D) 39.1%
North Carolina 7	Mike McIntyre	1996	Democratic	Mike McIntyre (D) 50.1% David Rouzer (R) 49.9%
North Carolina 8	Larry Kissell	2008	Democratic	Richard Hudson (R) 54.1% Larry Kissell (D) 45.9%
North Carolina 10	Patrick McHenry	2004	Republican	Patrick McHenry (R) 57.0% Patsy Keever (D) 43.0%

 Who were the candidates for the race in North Carolina's 4th district in 2012? And summarize the result of that election.

 David Price from Democratic and Tim D'Annunzio from Republican were candidates for the race in North Carolina's 4th district in 2012. David was re-elected with a wide margin, receiving 74.4% of the vote compared to Tim's 25.6%. This election result showed a clear victory for David, who had been in this position since 1996.

 According to the voting result, which representative election in North Carolina districts was the most competitive, and why?

 The race in the North Carolina 7th district was the most competitive, as the Democratic incumbent Mike McIntyre won by a slim margin, with only a 0.2% difference between him and his Republican challenger David Rouzer. Furthermore, this election was the only one among all North Carolina districts in 2012 that resulted in a margin of victory within less than 1 percent.

QTSumm

	in-domain	out-of-domain		
	MultiNLI	SNLI	Glockner	SICK
MQAN	72.30	60.91	41.82	53.95
+ coverage	73.84	65.38	78.69	54.55
ESIM (ELMO)	80.04	68.70	60.21	51.37
+ coverage	80.38	70.05	67.47	52.65

Table 2: Impact of using coverage for improving generalization across different datasets of the same task (NLI). All models are trained on MultiNLI.

Table 2 shows the performance for both systems for in-domain (the MultiNLI development set) as well as out-of-domain evaluations on SNLI, Glockner, and SICK datasets.

The results show that coverage information considerably improves the generalization of both examined models across various NLI datasets. The resulting cross-dataset improvements on the SNLI and Glockner datasets are larger than those on the SICK dataset. The reason is that the dataset creation process and therefore, the task formulation is similar in SNLI and MultiNLI, but they are different from SICK. In particular, in the neutral pairs

SciGen

- 1) Zhao et al., QTSumm: Query-Focused Summarization over Tabular Data, EMNLP 2023
- 2) Moosavi et al., SciGen: a Dataset for Reasoning-Aware Text Generation from Scientific Tables, NeurIPS 2021

Prompting-based In-Context Learning

- Investigating the capabilities of LLMs in T2T for three aspects
 - LLM's faithfulness is compared to existing methods
 - $Y = LLM(prompt(Table))$
 - LLMs are used as an evaluator to verify generated text (CoT-Acc)
 - $Y_{T/F} = LLM(prompt(Table, \hat{Y}))$
 - LLMs are used to give feedback (correction) to fine-tuned models

Title: 1964 United States Presidential Election in Illinois

Party	Candidates	Votes	Votes %
Democratic	Lyndon B. Johnson	2,796,833	59.47%
Republican	Barry Goldwater (... abbreviation...)	1,905,946	40.53%

Information Seeking Scenario 1: Data Insight Generation

Here are some meaningful insights from the given table about the 1964 United States presidential election in Illinois:

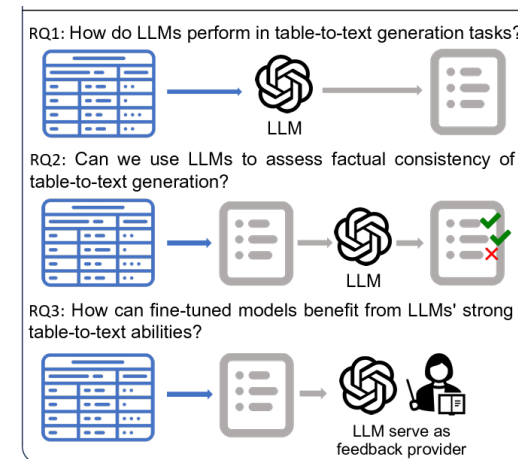
1.
2.

Information Seeking Scenario 2: Query-based Generation

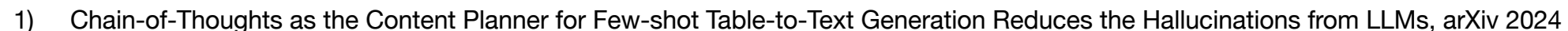
How did Lyndon B. Johnson fare against his opponent in the Illinois presidential election?

Lyndon B. Johnson won Illinois with 59.47% of the vote, against Barry Goldwater, who received 40.53% of the vote.

Metric	Acc on Tabfact	Pearson's correlation
SP-Acc	63.5	.458
NLI-Acc	65.1	.526
TAPAS-Acc	81.0	.705
TAPEX-Acc	84.2	.804
CoT-3.5-Acc	78.0	.787
CoT-4-Acc	80.9	.816

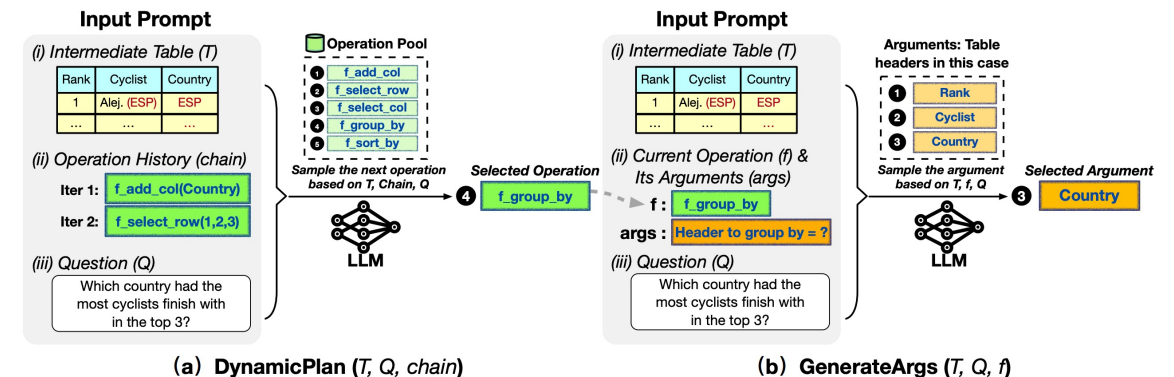
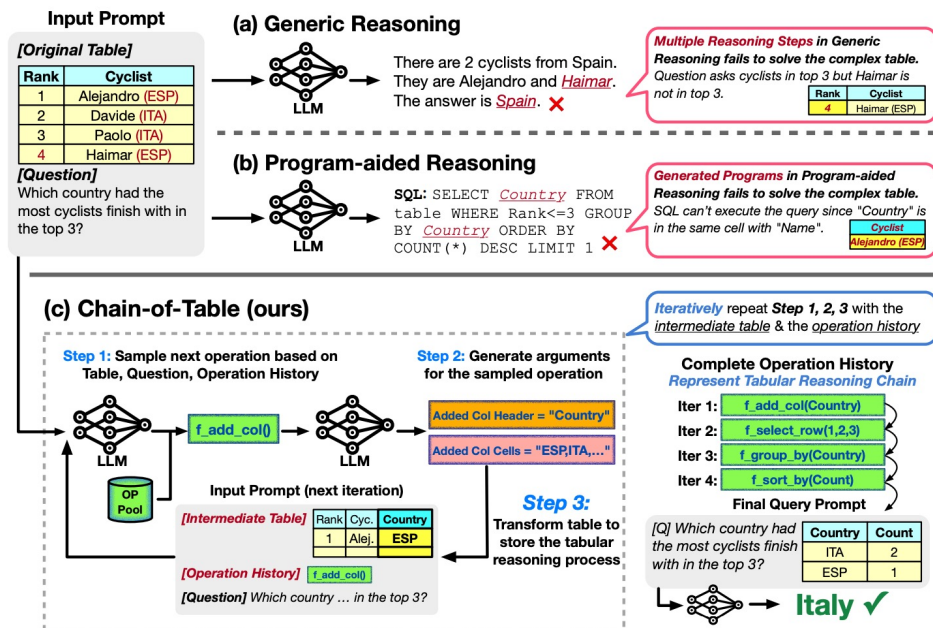


- CoT-Planner uses the reasoning ability of LLMs for planning and target generation
 - It first rationalizes the 10 different candidate plans by feeding LLM both target and table
 - Then, it gives the content plan as the demonstration (shot) example



Prompting-based In-Context Learning

• Chain-of-Table



• Method

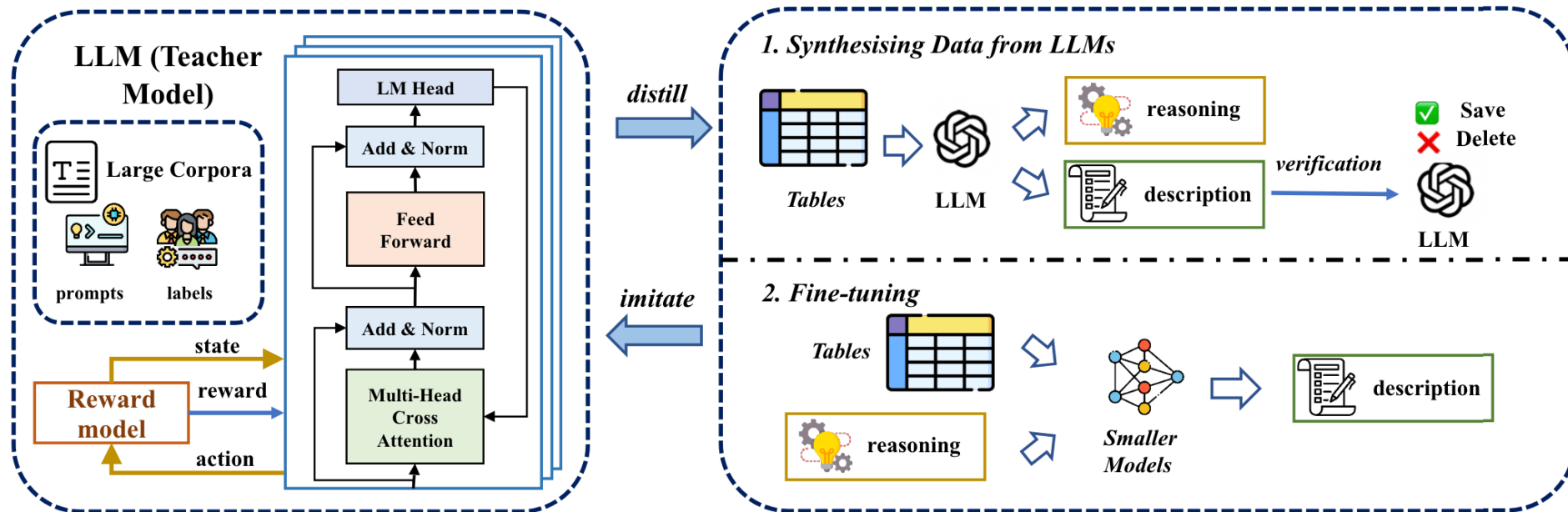
- Show examples of Operation → New Table pairs.
- At each step, inspect the latest table, apply the next operation, emit the updated table.
- Read the required value from the final table for the answer

• Motivation

- Text-only reasoning hides table structure → errors.
- Let the model **build intermediate tables**, like SQL → clearer, more accurate answers.

Reasoning Distillation from Teacher

- LLM is used as a teacher model to generate the reasoning chain required for target text
- The teacher model's reasoning ability is distilled to a smaller LM



Reasoning Distillation from Teacher

- HeLM uses a **highlighted table** as **evidence** for query-focused table summarization
 - It generates evidence data using a teacher model

$$E = \mathcal{M}_H (\text{Prompt}_H(T, Q))$$

E = evidence on table to generate target text

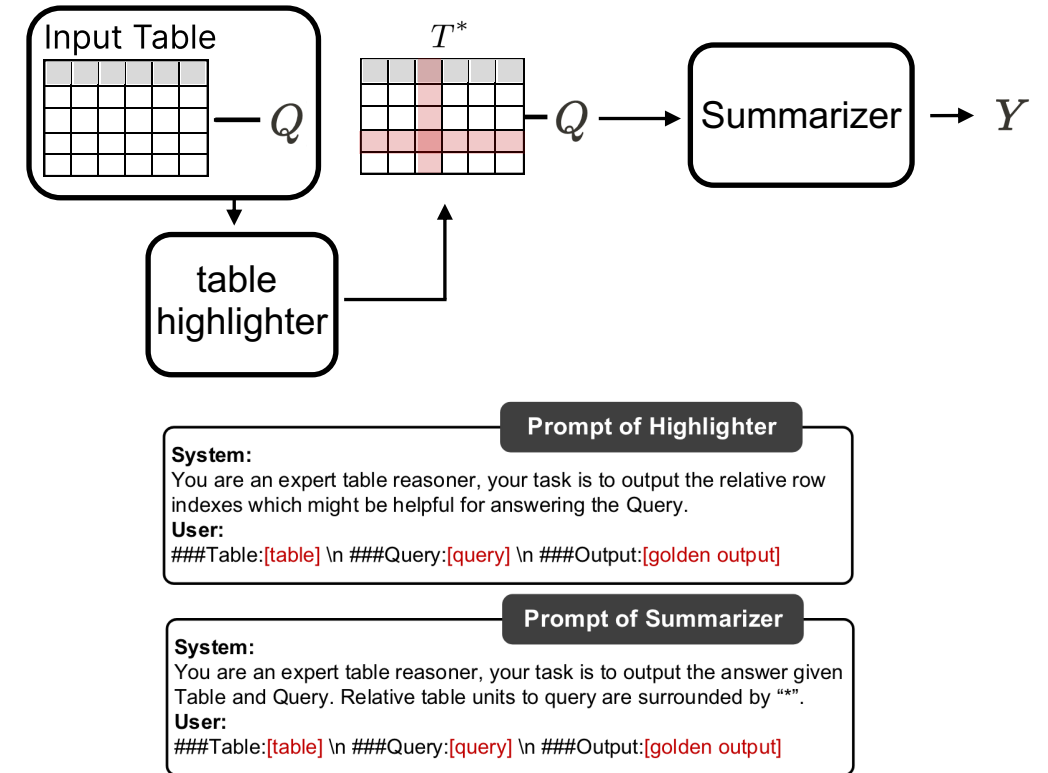
$$T^* = \text{HL}(T, E)$$

T^* = input table highlighted with evidence

- It trains a student model with generated data

$$\mathcal{L}_{evidence} = \frac{1}{N} \sum_{n=1}^N \log P(E|T, Q)$$

$$\mathcal{L}_{summary} = \frac{1}{N} \sum_{n=1}^N \log P(Y|T^*, Q)$$



Reasoning Distillation from Teacher

• QtP: Question-then-Point

Title: Kyle's Football Career Statistics

Season	Team	...	Injury	Goals
2003	A.Utd		X	2
2004	A.Utd		O	0
2005	A.Utd		X	1
2006	A.Utd		O	0
2007	B.FC		O	3
2008	B.FC		X	9
2009	B.FC		X	7
2010	B.FC		O	2
2011	C.City		O	0
2012	C.City		X	3

Implicit Table Knowledge

Target Summary

... Despite a brief successful season after the team transfer, his performance remain inconsistent and it seems that his injury impacted his goal performance throughout his career. ...

Implicit Table Knowledge

The table is about career statistics of a football player. Based on title and header, "player's performance trends" and "factors affecting performance" are important aspect from the table.

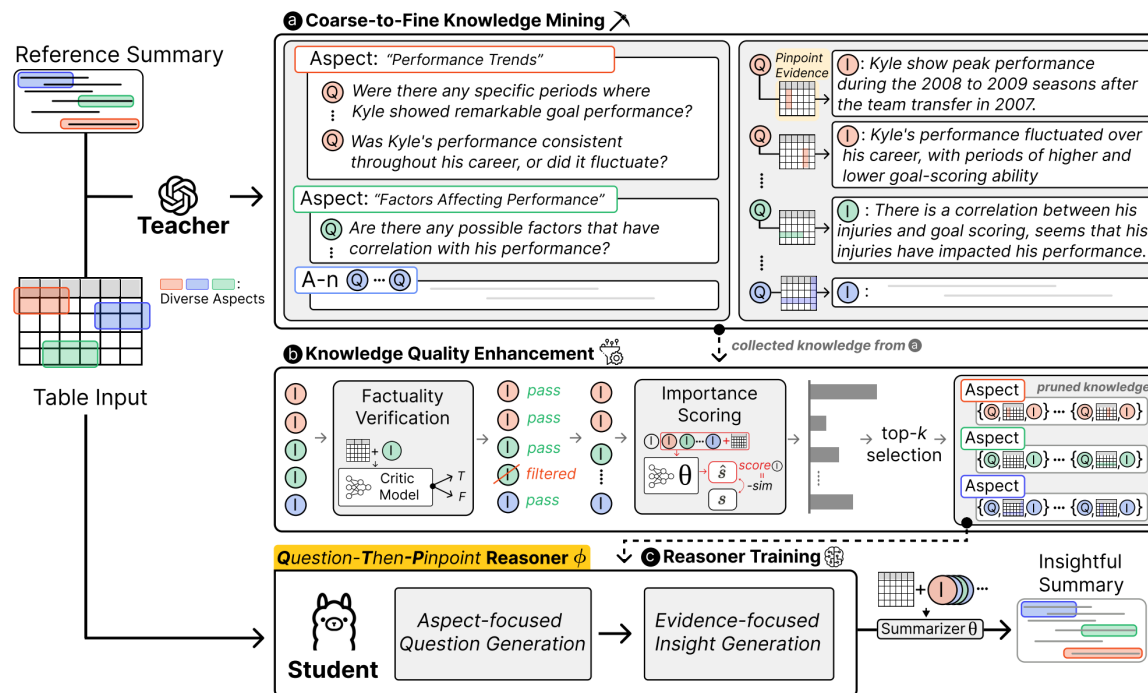
■ : coarse-level aspects

Observing the fluctuating numbers of "Goals" column, there are some significant variations with respect to the teams and injury.

There are some correlation between the injury and his goal performance.

After moving to another team, his goal scoring performance get peak during the 2008-09 season

■ ■ ■ : fine-level insights



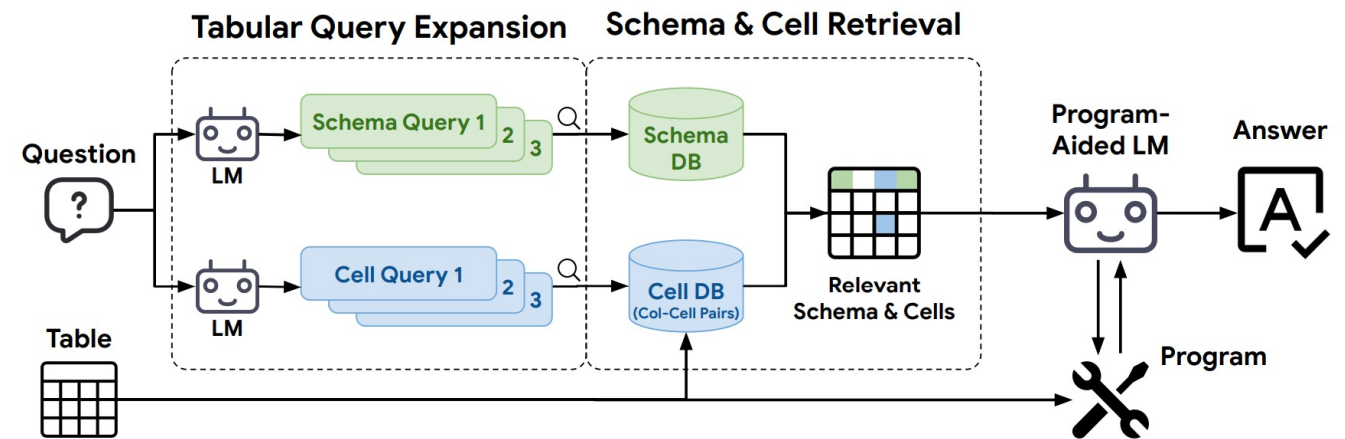
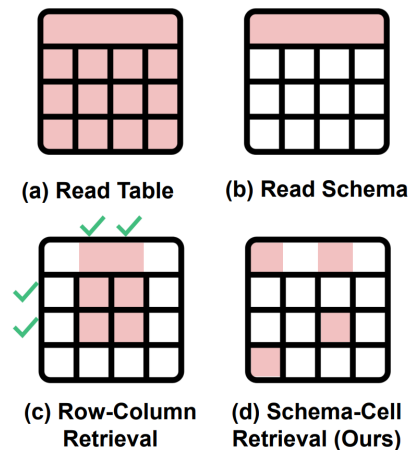
$$\mathcal{A}, \mathcal{Q} = LLM_{teacher}(t, s) \quad \mathcal{L}_{QG}(t, \mathcal{A}, \mathcal{Q}) = -\log p_{\phi}(\mathcal{A}, \mathcal{Q}|t)$$

$$\mathcal{E}, \mathcal{I} = LLM_{teacher}(t, s, \mathcal{Q}) \quad \mathcal{L}_{IG}(t, \mathcal{Q}, \mathcal{E}, \mathcal{I}) = -\sum_{(q, e, i) \in (\mathcal{Q}, \mathcal{E}, \mathcal{I})} \log p_{\phi}(e, i|t, q)$$

$$\mathcal{L}_{Reasoner} = \mathbb{E}_{(t, s, (\mathcal{A}, \mathcal{Q}, \mathcal{E}, \mathcal{I})) \sim D'} [\mathcal{L}_{QG}(t, \mathcal{A}, \mathcal{Q}) + \mathcal{L}_{IG}(t, \mathcal{Q}, \mathcal{E}, \mathcal{I})]$$

Retrieval Augmented Generation from Table

- **TableRAG - RAG from the Large single table**



- **Motivation**

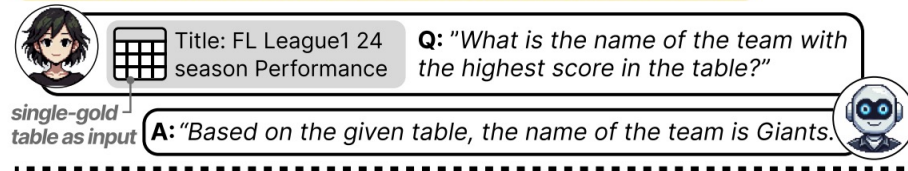
- Real-world tables can run into millions of tokens, far beyond any LLM context window; truncating them drops evidence and hurts accuracy.
- If we can **retrieve only the relevant pieces from the given table**, LLMs stay within budget yet see all relevant facts.

- **Query expansion** – enrich the user question with column names & keywords.
- **Two-stage retrieval**
 - ✓ *Schema retriever* picks the most relevant headers.
 - ✓ *Cell retriever* pulls the top rows/cells inside those headers.

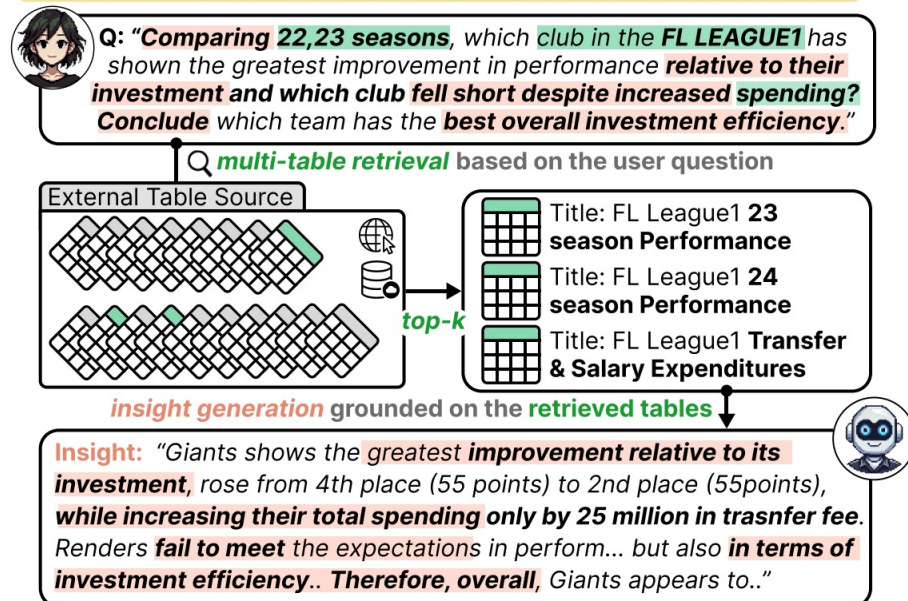
Retrieval Augmented Generation from Table

- MT-RAIG - RAG from External Table DB

Conventional *factoid-level* Table Question Answering



Retrieval-augmented *Insight* Generation over Multiple Tables



- Problem formulation

$$\hat{T}_q = Ret(q, T), \quad i = Gen(q, \hat{T}_q)$$

- Challenges

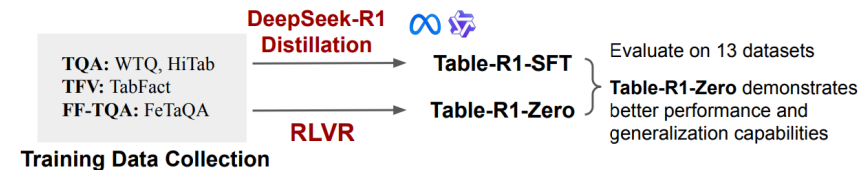
- retrieve query-relevant tables from the data store
- faithfully extract evidences scattered among irrelevant tables
- comprehensively aggregate these evidences to derive meaningful data insights

RL & Test-time Scaling for Table Reasoning

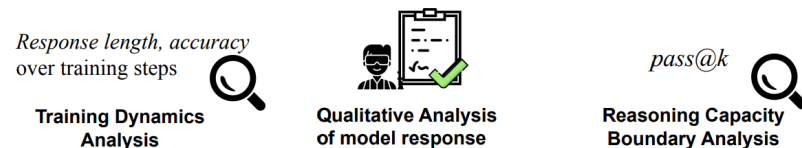
- **Table-R1**

RQ1: How effective is inference-time scaling for table reasoning tasks?

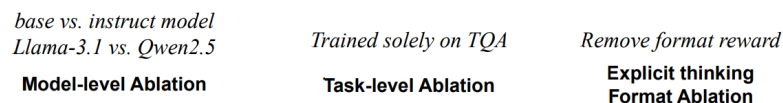
1. Evaluate performance of general-purpose reasoning LLMs
2. Explore two model training approaches



RQ2: What contributes to the success of RLVR methods in table reasoning tasks?



RQ3: How do various factors influence the effectiveness of RLVR training in table reasoning tasks?



Motivation

- Table reasoning adds structure, arithmetic and alignment challenges that frontier models still struggle
- Generating long reasoning traces at test also helps table tasks
- Aim for small open models to rival frontier LLMs on diverse table-QA, verification and free-form tasks

RQ1 — Inference-time scaling

→ Allowing a 7 B Table-R1 model to “think out loud” (with SFT or RLVR) lifts it to *GPT-4-level* accuracy

RQ2 — Why RLVR matters

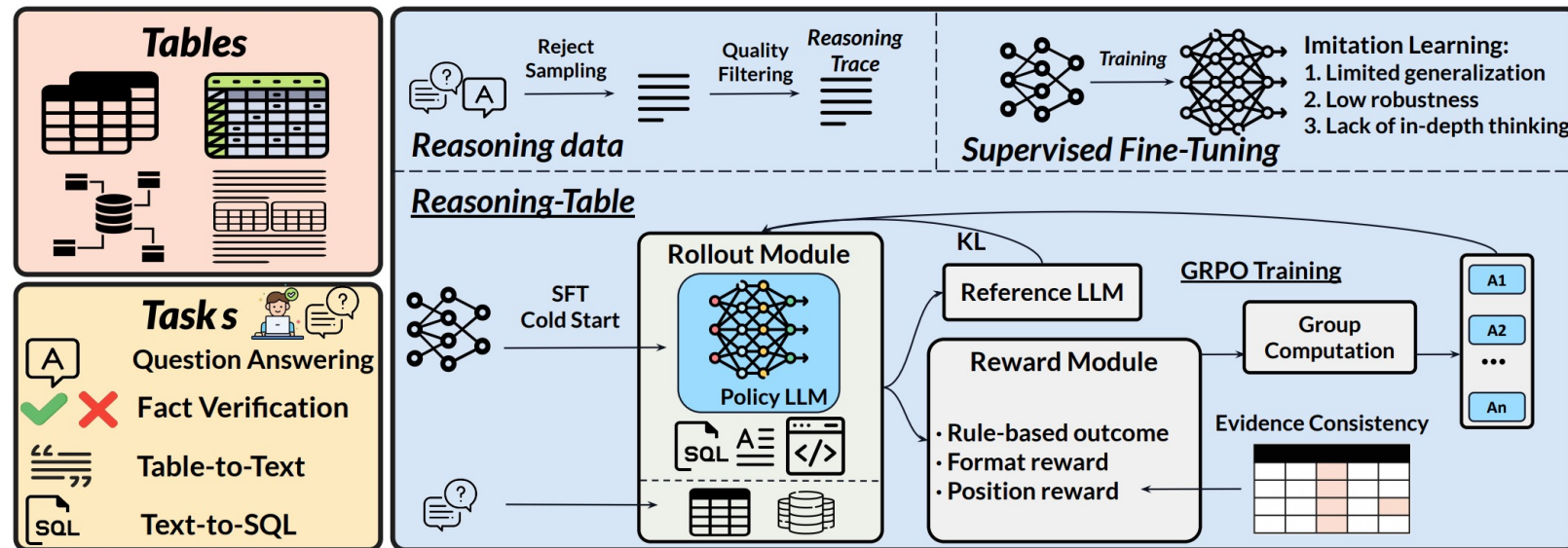
→ Verifiable, format-aware rewards force the model to produce clean step-by-step chains, sharply boosting exact-answer accuracy and pass@k.

RQ3 — What shapes RLVR’s effectiveness?.

- *Data scope*: TQA-only training transfers well to similar tasks, less so to free-form QA.
- *Rewards*: removing the chain/format reward destabilises training and slashes performance.

RL & Test-time Scaling for Table Reasoning

- Reasoning-Table



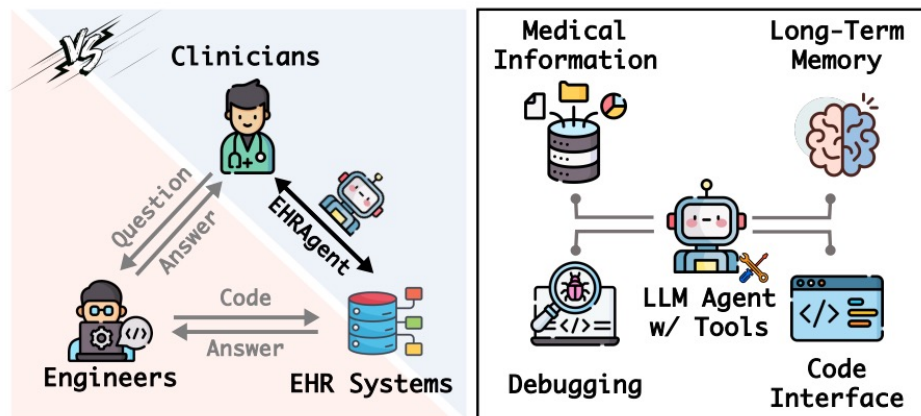
- Reward Design:

- Outcome reward** = correct answer / exec result / BLEU / F1 (Diverse Table Tasks)
- Format reward** = output must contain <think> & <answer> tags
- Position reward** = tag the true columns / cells used in reasoning (cell pinpointing)

→ different rewards weighted into a single scalar $R = R_{\text{out}} + \alpha R_{\text{fmt}} + \beta R_{\text{out}} \times R_{\text{pos}}$

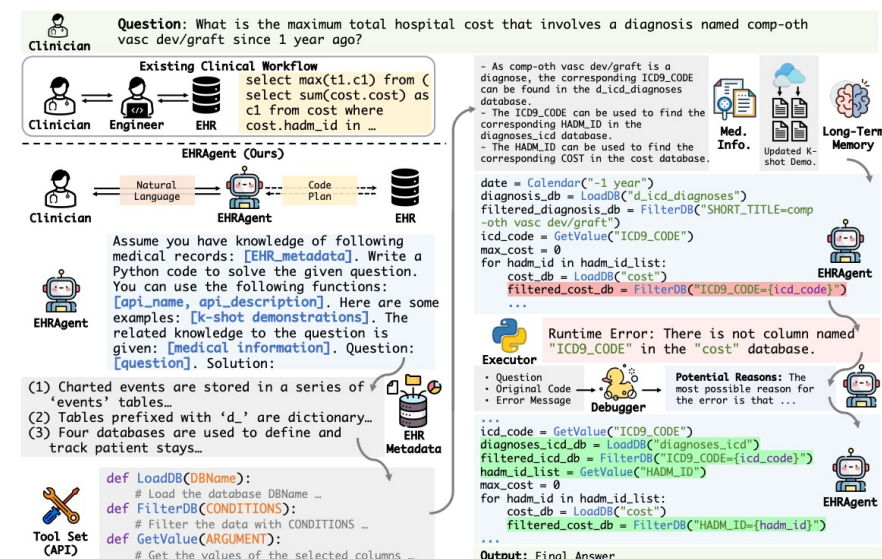
Agents for Table

- **EHR Agent**



- **Motivation**

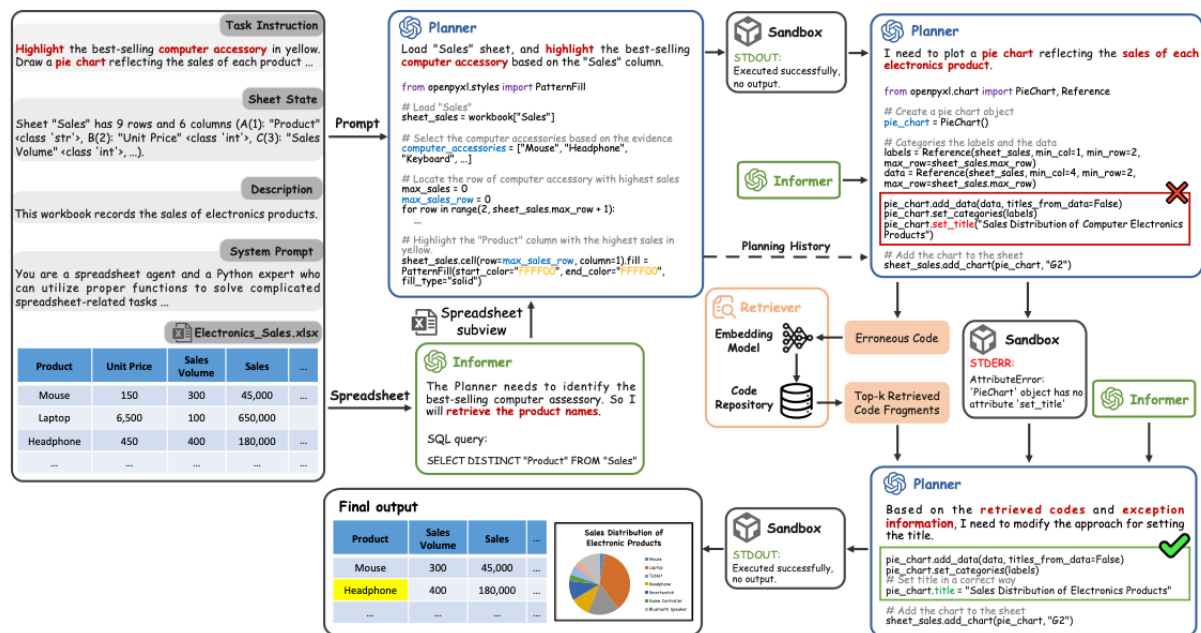
- answering clinical questions often means **joining, filtering, and aggregating** across 10+ related tables
- treat the task **as tool-use planning**—let an LLM write & debug short Python code (pandas) to query the table



- **Plan → Code → Execute loop**
The LLM proposes code, executes it, reads errors/outputs, then self-edits until a TERMINATE flag or step limit
- **Interactive debugging + memory**
memory of past successful snippets; retrieves the most relevant one to bootstrap new problems.
- **Plug-and-play tool set & domain hints**
Medical-specific tool wrappers (date ranges, unit conversion, risk scores) are exposed in natural language so the model can call them.

Agents for Table

• SheetAgent

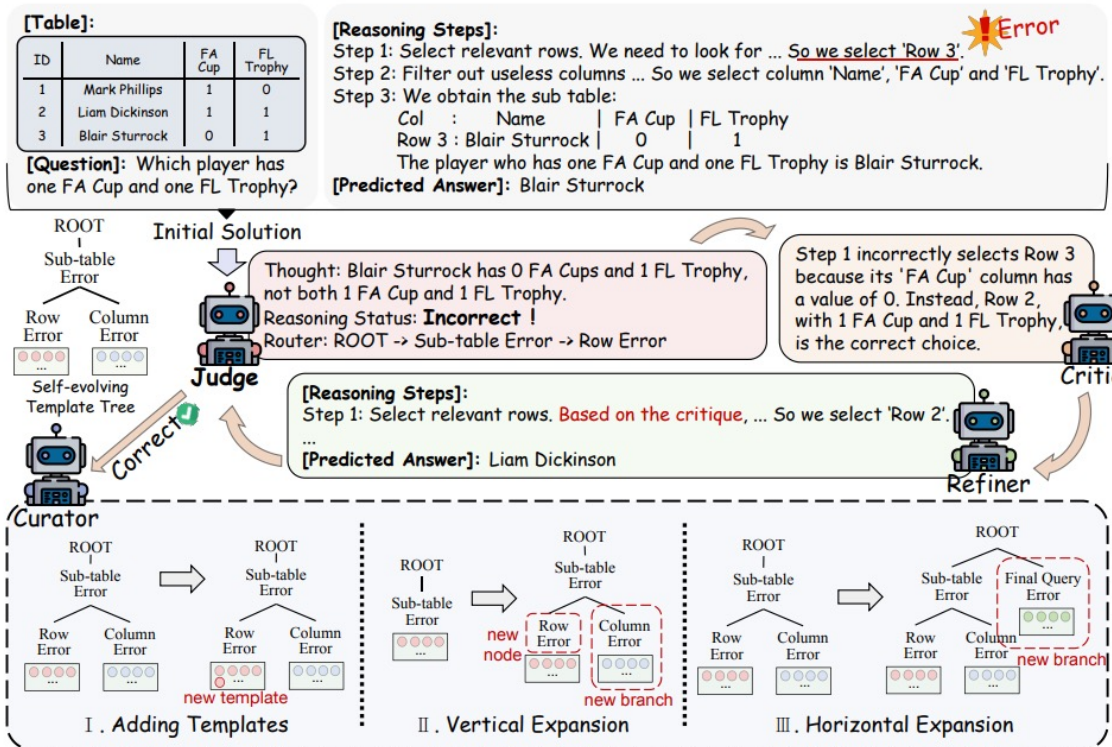


- **Planner** – decomposes the user query into an ordered list of atomic actions (filter, formula, join, chart, ...).
- **Informer** – executes each action via a Python/Excel API, checks results or error traces, and reflects if needed.
- **Retriever** – supplies the Planner with fresh sheet snapshots and task-relevant rows/columns.

- **Real-life spreadsheets are messy:** long-horizon, multi-table tasks with ambiguous requests, hard to solve with plain CoT or text-to-SQL models

Agents for Table

• Table-Critic



• Motivation

- Multi-step table reasoning often **propagates hidden errors**; single-LLM “self-reflection” is unreliable.
- Need to **spot & fix wrong intermediate steps** before they snowball

• Four specialist agents work in a loop:

- Judge** → flags the first flawed step.
- Critic** → writes a detailed, templated critique.
- Refiner** → rewrites the chain using the critique.
- Curator** → stores the critique in a **self-evolving template tree** so future cases reuse proven fixes.

-> Iterate until the Judge sees no error or a step cap is hit